

Complex Functions

Type I: Analytic function

1. Determine whether the following functions are analytic and if so find their derivatives

a. e^z

Ans. e^z

b. z^3

Ans. $3z^2$

c. $\sinh z$

[M15/CompIT/5M][M17/CompIT/5M][N17/Comp/5M]

Solution:

Let $f(z) = \sinh z$

$$\begin{aligned} f(z) &= \sinh(x + iy) \\ &= \sinh x \cdot \cosh iy + \cosh x \cdot \sinh iy \\ &= \sinh x \cos y + i \cosh x \sin y \end{aligned}$$

Now,

$$\begin{aligned} u &= \sinh x \cos y; & v &= \cosh x \sin y \\ u_x &= \cos y [\cosh x] & v_x &= \sin y [\sinh x] \\ u_y &= \sinh x [-\sin y] & v_y &= \cosh x [\cos y] \end{aligned}$$

We see that, $u_x = v_y$ & $u_y = -v_x$

$\therefore f(z) = \sinh z$ is analytic

And,

$$f(z) = u + iv$$

Diff w.r.t x , we get

$$f'(z) = u_x + i v_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + i v_{x(z,0)}$$

$$f'(z) = \cos 0 [\cosh z] + i \sin 0 [\sinh z]$$

$$\therefore f'(z) = \cosh z$$

d. $\sin z$

Ans. $\cos z$

e. ze^z

[M16/AutoMechCivil/5M]

Solution:

Let $f(z) = ze^z$

$$\begin{aligned} f(z) &= (x + iy)e^{x+iy} \\ &= (x + iy)e^x e^{iy} \\ &= e^x (x + iy)(\cos y + i \sin y) \\ &= e^x (x \cos y + ix \sin y + iy \cos y + i^2 y \sin y) \end{aligned}$$



$$= e^x[(x \cos y - y \sin y) + i(x \sin y + y \cos y)]$$

$$= e^x(x \cos y - y \sin y) + i e^x(x \sin y + y \cos y)$$

Now, $u = e^x x \cos y - e^x y \sin y$

$$u_x = \cos y [e^x(1) + x(e^x)] - y \sin y [e^x]$$

$$u_x = e^x [\cos y + x \cos y - y \sin y] \dots\dots\dots(1)$$

$$u_y = e^x x [-\sin y] - e^x [y(\cos y) + \sin y(1)]$$

$$u_y = e^x [-x \sin y - y \cos y - \sin y] \dots\dots\dots(2)$$

Also, $v = e^x x \sin y + e^x y \cos y$

$$v_x = \sin y [e^x(1) + x(e^x)] + y \cos y [e^x]$$

$$v_x = e^x [\sin y + x \sin y + y \cos y] \dots\dots\dots(3)$$

$$v_y = e^x x [\cos y] + e^x [y(-\sin y) + \cos y(1)]$$

$$v_y = e^x [x \cos y - y \sin y + \cos y] \dots\dots\dots(4)$$

We see that,

From (1) & (4), $u_x = v_y$

From (2) & (3), $u_y = -v_x$

$\therefore f(z) = ze^z$ is analytic

And,

$$f(z) = u + iv$$

Diff w.r.t x, we get

$$f'(z) = u_x + i v_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + i v_{x(z,0)}$$

$$f'(z) = e^z [\cos 0 + z \cos 0 - 0 \sin 0] + i e^z [\sin 0 + z \sin 0 + 0 \cos 0]$$

$$\therefore f'(z) = e^z [1 + z]$$

2. Prove that $f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \left(\frac{y}{x} \right)$ is analytic and find $f'(z)$.

$$\text{Ans. } \frac{1}{z}$$

3. Show that $f(z) = \frac{\bar{z}}{|z|^2}$, $|z| \neq 0$ is analytic function. Hence find $f'(z)$

$$\text{Ans. } -\frac{1}{z^2}$$

4. Is $f(z) = \frac{z}{\bar{z}}$ analytic?

[N16/Compt/5M]

Solution:

We have,

$$f(z) = \frac{z}{\bar{z}}$$

$$f(z) = \frac{r e^{i\theta}}{r e^{-i\theta}}$$

$$f(z) = e^{2i\theta}$$

$$f(z) = \cos 2\theta + i \sin 2\theta$$



$$\begin{aligned}\therefore u &= \cos 2\theta & v &= \sin 2\theta \\ u_r &= 0 & v_r &= 0 \\ u_\theta &= -2\sin 2\theta & v_\theta &= 2\cos 2\theta\end{aligned}$$

We see that,

$$\begin{aligned}ru_r &\neq v_\theta \\ rv_r &\neq -u_\theta \\ \therefore f(z) = \frac{z}{\bar{z}} &\text{ is not analytic}\end{aligned}$$

5. Determine whether the following functions are analytic and if so find their derivatives

a. $(x^3 - 3xy^2 + 2xy) + i(3x^2y - x^2 + y^2 - y^3)$ Ans. $3z^2 - 2iz$

b. $(x^3 + 3xy^2 - 3x) + i(3x^2y - y^3 + 3y)$

[M18/Comp/5M]

Solution:

$$f(z) = x^3 + 3xy^2 - 3x + i(3x^2y - y^3 + 3y)$$

Now,

$$\begin{aligned}u &= x^3 + 3xy^2 - 3x; & v &= (3x^2y - y^3 + 3y) \\ u_x &= 3x^2 + 3y^2 - 3 & v_x &= 6xy \\ u_y &= 6xy & v_y &= 3x^2 - 3y^2 + 3\end{aligned}$$

We see that, $u_x \neq v_y$ & $u_y \neq -v_x$

$\therefore f(z)$ is not analytic

c. $\cosh z$

Ans. $\sinh z$

d. e^{2z}

Ans. $2e^{2z}$

e. z^n

[M19/Biom/5M][M19/Elex/5M]

Solution:

$$\text{Let } z = re^{i\theta}, \therefore z^n = r^n e^{in\theta}$$

$$\therefore z^n = r^n (\cos n\theta + i \sin n\theta)$$

$$u = r^n \cos n\theta, v = r^n \sin n\theta$$

$$u_r = nr^{n-1} \cos n\theta, u_\theta = -r^n \cdot n \sin n\theta$$

$$v_r = nr^{n-1} \sin n\theta, v_\theta = nr^n \cos n\theta$$

$$\therefore ru_r = v_\theta \text{ \& } rv_r = -u_\theta$$

Thus, $f(z) = z^n$ is analytic.

$$f'(z) = e^{-i\theta} [u_r + i v_r]$$

$$f'(z) = e^{-i\theta} [nr^{n-1} \cos n\theta + i nr^{n-1} \sin n\theta]$$

$$f'(z) = nr^{n-1} e^{-i\theta} [\cos n\theta + i \sin n\theta]$$

$$f'(z) = nr^{n-1} \cdot e^{-i\theta} \cdot e^{in\theta}$$

$$f'(z) = nr^{n-1} \cdot e^{i(n-1)\theta}$$

$$f'(z) = n(re^{i\theta})^{n-1} = nz^{n-1}$$

OR



$$f'(z) = e^{-i\theta}[u_r + iv_r]$$

By Milne Thomson Method

$$f'(z) = \{e^{-i\theta}[u_r + iv_r]\}_{(z,0)}$$

$$f'(z) = \{e^{-i\theta}[nr^{n-1}\cos n\theta + inr^{n-1}\sin n\theta]\}_{(z,0)}$$

$$f'(z) = e^0[nz^{n-1}\cos 0 + inz^{n-1}\sin 0]$$

$$f'(z) = nz^{n-1}$$

f. $\bar{z}$

Ans. not analytic

g. $\log z$

Ans. $\frac{1}{z}$

h. $x^2 - y^2 + 2ixy$

[N14/ElexExtcElectBiomInst/5M][M19/Comp/5M]

Solution:

Let $f(z) = x^2 - y^2 + 2ixy$

Now,

$$u = x^2 - y^2$$

$$v = 2xy$$

$$u_x = [2x] - 0$$

$$v_x = 2y[1]$$

$$u_y = 0 - [2y]$$

$$v_y = 2x[1]$$

We see that, $u_x = v_y$ & $u_y = -v_x$

$\therefore f(z)$ is analytic

And,

$$f(z) = u + iv$$

Diff w.r.t x, we get

$$f'(z) = u_x + i v_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + i v_{x(z,0)}$$

$$f'(z) = 2z + i 2(0)$$

$$\therefore f'(z) = 2z$$

6. Prove that $w = \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2}$ is analytic, find $\frac{dw}{dz}$ Ans. $-\frac{1}{z^2}$

7. Show that the function $f(z) = \frac{1}{r^2}(\cos 2\theta - i \sin 2\theta)$ is analytic

[N19/Chem/5M]

Solution:

$$f(z) = \frac{1}{r^2}(\cos 2\theta - i \sin 2\theta)$$

$$f(z) = \frac{1}{r^2}\cos 2\theta - i \frac{1}{r^2}\sin 2\theta$$

$$u = \frac{1}{r^2}\cos 2\theta, v = -\frac{1}{r^2}\sin 2\theta$$

$$u_r = \left[-\frac{2}{r^3}\right]\cos 2\theta, u_\theta = \frac{1}{r^2}[-2\sin 2\theta]$$



$$v_r = \left[\frac{2}{r^3} \right] \sin 2\theta, v_\theta = -\frac{1}{r^2} [2 \cos 2\theta]$$

$$\therefore ru_r = v_\theta \& rv_r = -u_\theta$$

Thus, $f(z) = z^n$ is analytic.

$$f'(z) = e^{-i\theta} [u_r + iv_r]$$

$$f'(z) = e^{-i\theta} \left[-\frac{2}{r^3} \cos 2\theta + \frac{i2}{r^3} \sin 2\theta \right]$$

$$f'(z) = -\frac{2}{r^3} e^{-i\theta} [\cos 2\theta - i \sin 2\theta]$$

$$f'(z) = -\frac{2}{r^3} \cdot e^{-i\theta} \cdot e^{-2i\theta}$$

$$f'(z) = -\frac{2}{r^3} \cdot e^{-3i\theta}$$

$$f'(z) = -\frac{2}{r^3 e^{3i\theta}}$$

$$f'(z) = -\frac{2}{(re^{i\theta})^3}$$

$$f'(z) = -\frac{2}{z^3}$$

OR

$$f'(z) = \{e^{-i\theta} [u_r + iv_r]\}_{(z,0)}$$

$$f'(z) = \left\{ e^{-i\theta} \left[-\frac{2}{r^3} \cos 2\theta + \frac{i2}{r^3} \sin 2\theta \right] \right\}_{(z,0)}$$

$$f'(z) = e^0 \left[-\frac{2}{z^3} \cos 0 + \frac{2i}{z^3} \sin 0 \right]$$

$$f'(z) = -\frac{2}{z^3}$$

8. Find k such that $\frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \frac{kx}{y}$ is analytic.

[M15/ChemBiot/5M][N16/ChemBiot/5M][N16/AutoMechCivil/6M]

[N17/IT/6M][M19/Chem/5M][N19/IT/6M]

Solution:

We have,

$$f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \frac{kx}{y}$$

$$u = \frac{1}{2} \log(x^2 + y^2)$$

$$u_x = \frac{1}{2} \left[\frac{1}{x^2 + y^2} \times 2x \right]$$

$$u_y = \frac{1}{2} \left[\frac{1}{x^2 + y^2} \times 2y \right]$$

$$v = \tan^{-1} \frac{kx}{y}$$

$$v_x = \frac{1}{1 + \left(\frac{kx}{y}\right)^2} \times \frac{k}{y} \quad (1)$$

$$v_y = \frac{1}{1 + \left(\frac{kx}{y}\right)^2} \times kx \left(-\frac{1}{y^2} \right)$$

Since, $f(z)$ is given analytic, we know that

$$\begin{aligned} u_x &= v_y \\ \frac{1}{2} \left[\frac{2x}{x^2 + y^2} \right] &= \frac{1}{1 + \frac{k^2 x^2}{y^2}} \times -\frac{kx}{y^2} \end{aligned}$$



$$\frac{x}{x^2+y^2} = \frac{y^2}{y^2+k^2x^2} \times -\frac{kx}{y^2}$$

$$\frac{x}{x^2+y^2} = -\frac{kx}{k^2x^2+y^2}$$

On Comparison, we get

$$\boxed{k = -1}$$

9. Find the constants a, b, c, d, e If

$f(z) = (ax^4 + bx^2y^2 + cy^4 + dx^2 - 2y^2) + i(4x^3y - exy^3 + 4xy)$
is analytic.

[N13/CompIT/5M][M14/ChemBiot/5M][N15/CompIT/5M]

[N16/ElexExtcElectBiomInst/5M][N18/IT/5M]

Solution:

$$f(z) = (ax^4 + bx^2y^2 + cy^4 + dx^2 - 2y^2) + i(4x^3y - exy^3 + 4xy)$$

$$u = ax^4 + bx^2y^2 + cy^4 + dx^2 - 2y^2$$

$$u_x = a[4x^3] + by^2[2x] + 0 + d[2x] - 0$$

$$u_y = 0 + bx^2[2y] + c[4y^3] + 0 - 2[2y]$$

$$v = 4x^3y - exy^3 + 4xy$$

$$v_x = 4y[3x^2] - ey^3[1] + 4y[1]$$

$$v_y = 4x^3[1] - ex[3y^2] + 4x[1]$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$4ax^3 + 2bxy^2 + 2dx = 4x^3 - 3exy^2 + 4x$$

Comparing the coefficients, we get

$$4a = 4 \quad \& \quad 2b = -3e \quad \& \quad 2d = 4$$

$$\boxed{a = 1} \quad \& \quad 2b + 3e = 0 \dots\dots(1) \quad \& \quad \boxed{d = 2}$$

Also,

$$u_y = -v_x$$

$$2bx^2y + 4cy^3 - 4y = -12x^2y + ey^3 - 4y$$

Comparing the coefficients, we get

$$2b = -12 \quad \& \quad 4c = e \dots\dots(2)$$

$$\boxed{b = -6}$$

From (1),

$$2b + 3e = 0,$$

$$2(-6) + 3e = 0$$

$$\therefore \boxed{e = 4}$$

From (2),

$$4c = e$$

$$\therefore \boxed{c = 1}$$

10. Find p if $f(z) = r^2 \cos 2\theta + i r^2 \sin p\theta$ is analytic
[M17/AutMechCivil/5M][M18/Elect/5M][M18/MTRX/5M]

Solution:

We have,

$$f(z) = r^2 \cos 2\theta + i r^2 \sin p\theta$$

$$u = r^2 \cos 2\theta$$

$$u_r = \cos 2\theta [2r]$$

$$u_\theta = r^2 [-\sin 2\theta \times 2]$$

$$v = r^2 \sin p\theta$$

$$v_r = \sin p\theta [2r]$$

$$v_\theta = r^2 [\cos p\theta \times p]$$

Since, $f(z)$ is given analytic, we know that

$$r u_r = v_\theta$$

$$r \cos 2\theta [2r] = r^2 [\cos p\theta \times p]$$

$$2r^2 \cos 2\theta = p r^2 \cos p\theta$$

On Comparison, we get

$$\boxed{p = 2}$$

11. Find p such that $\frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \frac{py}{x}$ is analytic.
[N15/ElexExtcElectBiomInst/5M][M18/Inst/5M]

Solution:

We have,

$$f(z) = \frac{1}{2} \log(x^2 + y^2) + i \tan^{-1} \frac{py}{x}$$

$$u = \frac{1}{2} \log(x^2 + y^2)$$

$$u_x = \frac{1}{2} \left[\frac{1}{x^2 + y^2} \times 2x \right]$$

$$u_y = \frac{1}{2} \left[\frac{1}{x^2 + y^2} \times 2y \right]$$

$$v = \tan^{-1} \frac{py}{x}$$

$$v_x = \frac{1}{1 + \left(\frac{py}{x}\right)^2} \times py \left(-\frac{1}{x^2}\right)$$

$$v_y = \frac{1}{1 + \left(\frac{py}{x}\right)^2} \times \frac{p}{x} (1)$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$\frac{1}{2} \left[\frac{2x}{x^2 + y^2} \right] = \frac{1}{1 + \frac{p^2 y^2}{x^2}} \times \frac{p}{x}$$

$$\frac{x}{x^2 + y^2} = \frac{x^2}{x^2 + p^2 y^2} \times \frac{p}{x}$$

$$\frac{x}{x^2 + y^2} = \frac{px}{x^2 + p^2 y^2}$$

On Comparison, we get

$$\boxed{p = 1}$$

12. Determine the constant a, b, c, d if
 $f(z) = x^2 + 2axy + by^2 + i(cx^2 + 2dxy + y^2)$ is analytic
[N13/AutoMechCivil/5M] [M15/ElexExtcElectBiomInst/5M]
[N17/Inst/5M][N19/Extc/5M]

Solution:

We have,

$$f(z) = x^2 + 2axy + by^2 + i(cx^2 + 2dxy + y^2)$$

$$u = x^2 + 2axy + by^2$$

$$v = cx^2 + 2dxy + y^2$$

$$u_x = [2x] + 2ay[1] + 0$$

$$v_x = c[2x] + 2dy[1] + 0$$

$$u_y = 0 + 2ax[1] + b[2y]$$

$$v_y = 0 + 2dx[1] + [2y]$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$2x + 2ay = 2dx + 2y$$

Comparing the coefficients, we get

$$2d = 2 \text{ \& } 2a = 2$$

$$\boxed{d = 1} \text{ \& } \boxed{a = 1}$$

Also,

$$u_y = -v_x$$

$$2ax + 2by = -2cx - 2dy$$

Comparing the coefficients, we get

$$2a = -2c \text{ \& } 2b = -2d$$

$$a = -c \text{ \& } b = -d$$

$$\boxed{c = -1} \text{ \& } \boxed{b = -1}$$

13. Determine the constant a, b, c, d
 if $f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$ is analytic.
[N17/Extc/5M][M18/IT/5M][M19/Extc/5M][N19/Comp/5M]

Solution:

$$f(z) = x^2 + axy + by^2 + i(cx^2 + dxy + y^2)$$

$$u = x^2 + axy + by^2$$

$$v = cx^2 + dxy + y^2$$

$$u_x = [2x] + ay[1] + 0$$

$$v_x = c[2x] + dy[1] + 0$$

$$u_y = 0 + ax[1] + b[2y]$$

$$v_y = 0 + dx[1] + [2y]$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$2x + ay = dx + 2y$$

Comparing the coefficients, we get

$$\boxed{d = 2} \text{ \& } \boxed{a = 2}$$

Also,

$$u_y = -v_x$$

$$ax + 2by = -2cx - dy$$



Comparing the coefficients, we get

$$a = -2c \quad \& \quad 2b = -d$$

$$2 = -2c \quad \& \quad 2b = -2$$

$$\boxed{c = -1} \quad \& \quad \boxed{b = -1}$$

14. Determine the constant a,b,c,d if

$$f(z) = (10x^2 + ay^2 + 8bxy) + i(5cxy + 7dx^2 + 2x^2) \text{ is analytic}$$

[N17/Elect/5M]

Solution:

We have,

$$f(z) = 10x^2 + ay^2 + 8bxy + i(5cxy + 7dx^2 + 2x^2)$$

$$u = 10x^2 + ay^2 + 8bxy \quad v = 5cxy + 7dx^2 + 2x^2$$

$$u_x = 10[2x] + 0 + 8by[1] \quad v_x = 5cy[1] + 7d[2x] + 4x$$

$$u_y = 0 + a[2y] + 8bx[1] \quad v_y = 5cx[1] + 0 + 0$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$20x + 8by = 5cx$$

Comparing the coefficients, we get

$$5c = 20 \quad \& \quad 8b = 0$$

$$\boxed{c = 4} \quad \& \quad \boxed{b = 0}$$

Also,

$$u_y = -v_x$$

$$2ay + 8bx = -5cy - 14dx - 4x$$

Comparing the coefficients, we get

$$2a = -5c \quad \& \quad 8b = -14d - 4$$

$$2a = -20 \quad \& \quad 0 = -14d - 4$$

$$\boxed{a = -10} \quad \& \quad \boxed{d = -\frac{2}{7}}$$

15. Determine the constant a,b,c,d if

$$f(z) = (4x^2 + ay^2 + 8bxy) + i(2cxy + 4dx^2 + 2x^2) \text{ is analytic}$$

[M19/Inst/5M]

Solution:

$$f(z) = (4x^2 + ay^2 + 8bxy) + i(2cxy + 4dx^2 + 2x^2)$$

$$u = 4x^2 + ay^2 + 8bxy \quad v = 2cxy + 4dx^2 + 2x^2$$

$$u_x = 4[2x] + 0 + 8by[1] \quad v_x = 2cy[1] + 4d[2x] + 4x$$

$$u_y = 0 + a[2y] + 8bx[1] \quad v_y = 2cx[1] + 0 + 0$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$8x + 8by = 2cx$$



Comparing the coefficients, we get

$$2c = 8 \quad \& \quad 8b = 0$$

$$\boxed{c = 4} \quad \& \quad \boxed{b = 0}$$

Also,

$$u_y = -v_x$$

$$2ay + 8bx = -2cy - 8dx - 4x$$

Comparing the coefficients, we get

$$2a = -2c \quad \& \quad 8b = -8d - 4$$

$$2a = -8 \quad \& \quad 0 = -8d - 4$$

$$\boxed{a = -4} \quad \& \quad \boxed{d = -\frac{1}{2}}$$

16. Find the constants a, b, c, d, e if

$$f(z) = (ax^3 + bxy^2 + 3x^2 + cy^2 + x) + i(dx^2y - 2y^3 + exy + y)$$

is analytic.

[M18/N18/Biom/5M][M19/AutoMechCivil/6M]

Solution:

We have,

$$f(z) = (ax^3 + bxy^2 + 3x^2 + cy^2 + x) + i(dx^2y - 2y^3 + exy + y)$$

$$u = ax^3 + bxy^2 + 3x^2 + cy^2 + x$$

$$u_x = a[3x^2] + by^2[1] + 3[2x] + 0 + 1$$

$$u_y = 0 + bx[2y] + 0 + c[2y] + 0$$

$$v = dx^2y - 2y^3 + exy + y$$

$$v_x = dy[2x] - 0 + ey[1] + 0$$

$$v_y = dx^2[1] - 2[3y^2] + ex[1] + 1$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$3ax^2 + by^2 + 6x + 1 = dx^2 - 6y^2 + ex + 1$$

Comparing the coefficients, we get

$$3a = d \quad \& \quad b = -6 \quad \& \quad 6 = e$$

$$3a = d \dots\dots (1) \quad \& \quad \boxed{b = -6} \quad \& \quad \boxed{e = 6}$$

Also, $u_y = -v_x$

$$2bxy + 2cy = -2dxy - ey$$

Comparing the coefficients, we get

$$2b = -2d \quad \& \quad 2c = -e$$

$$2(-6) = -2d \quad \& \quad 2c = -6$$

$$\boxed{d = 6} \quad \& \quad \boxed{c = -3}$$

From (1),

$$3a = d$$

$$3a = 6, \therefore \boxed{a = 2}$$



17. Find the constants a, b, c, d, e if

$$f(z) = (ax^3 + bxy^2 + 3x^2 + cy^2) + i(dx^2y - 2y^3 + exy)$$

is analytic.

[N18/Extc/5M]

Solution:

We have,

$$f(z) = (ax^3 + bxy^2 + 3x^2 + cy^2) + i(dx^2y - 2y^3 + exy)$$

$$u = ax^3 + bxy^2 + 3x^2 + cy^2$$

$$u_x = a[3x^2] + by^2[1] + 3[2x] + 0$$

$$u_y = 0 + bx[2y] + 0 + c[2y]$$

$$v = dx^2y - 2y^3 + exy$$

$$v_x = dy[2x] - 0 + ey[1]$$

$$v_y = dx^2[1] - 2[3y^2] + ex[1]$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$3ax^2 + by^2 + 6x = dx^2 - 6y^2 + ex$$

Comparing the coefficients, we get

$$3a = d \quad \& \quad b = -6 \quad \& \quad 6 = e$$

$$3a = d \dots\dots (1) \quad \& \quad \boxed{b = -6} \quad \& \quad \boxed{e = 6}$$

Also,

$$u_y = -v_x$$

$$2bxy + 2cy = -2dxy - ey$$

Comparing the coefficients, we get

$$2b = -2d \quad \& \quad 2c = -e$$

$$2(-6) = -2d \quad \& \quad 2c = -6$$

$$\boxed{d = 6} \quad \& \quad \boxed{c = -3}$$

From (1),

$$3a = d$$

$$3a = 6, \therefore \boxed{a = 2}$$

18. Determine the constants A, B, C, D, E & F if

$$f(z) = (Ax^3 - Bxy^2 + \sin 6x \cosh 6y + Cx) + i(Dyx^2 - 9y^3 + \cos Ex \sinh Fy + 101y)$$

[N19/Elect/5M]

Solution:

We have,

$$f(z) = (Ax^3 - Bxy^2 + \sin 6x \cosh 6y + Cx) + i(Dyx^2 - 9y^3 + \cos Ex \sinh Fy + 101y)$$

$$u = Ax^3 - Bxy^2 + \sin 6x \cosh 6y + Cx$$

$$u_x = A[3x^2] - By^2[1] + \cosh 6y [6\cos 6x] + C[1]$$



$$\begin{aligned}
 u_y &= 0 - Bx[2y] + \sin 6x [6 \sinh 6y] \\
 v &= Dx^2y - 9y^3 + \cos Ex \sinh Fy + 101y \\
 v_x &= Dy[2x] - 0 + \sinh Fy [-E \sin Ex] + 0 \\
 v_y &= Dx^2[1] - 9[3y^2] + \cos Ex [F \cosh Fy] + 101[1]
 \end{aligned}$$

Since, $f(z)$ is given analytic, we know that

$$u_x = v_y$$

$$3Ax^2 - By^2 + 6 \cos 6x \cosh 6y + C = Dx^2 - 27y^2 + F \cos Ex \cosh Fy + 101$$

Comparing the coefficients, we get

$$\begin{aligned}
 3A = D \quad & \& \quad -B = -27 \quad & \& \quad \boxed{F = 6} \quad & \& \quad \boxed{E = 6} \\
 3A = D \dots\dots (1) \quad & \& \quad \boxed{B = 27} \quad & \& \quad \boxed{C = 101}
 \end{aligned}$$

Also,

$$u_y = -v_x$$

$$-2Bxy + 6 \sin 6x \sinh 6y = -2Dxy + E \sin Ex \sinh Fy$$

Comparing the coefficients, we get

$$\begin{aligned}
 -2B = -2D \quad & \& \quad E = 6 \quad & \& \quad F = 6 \\
 -2(27) = -2d
 \end{aligned}$$

$$\boxed{D = 27}$$

From (1),

$$3A = D$$

$$3A = 27, \therefore \boxed{A = 9}$$

19. Determine constant m if $f(z) = r^7 \cos m\theta + ir^m \sin 7\theta$
[N18/Inst/5M]

Solution:

We have,

$$f(z) = r^7 \cos m\theta + ir^m \sin 7\theta$$

$$u = r^7 \cos m\theta$$

$$v = r^m \sin 7\theta$$

$$u_r = \cos m\theta [7r^6]$$

$$v_r = \sin 7\theta [mr^{m-1}]$$

$$u_\theta = r^7 [-\sin m\theta \times m]$$

$$v_\theta = r^m [\cos 7\theta \times 7]$$

Since, $f(z)$ is given analytic, we know that

$$ru_r = v_\theta$$

$$r \cos m\theta [7r^6] = r^m [\cos 7\theta \times 7]$$

$$7r^7 \cos m\theta = 7r^m \cos 7\theta$$

On Comparison, we get

$$\boxed{m = 7}$$

20. Determine constant m if $f(z) = r^5 \cos m\theta + ir^m \sin 5\theta$
[N17/Biom/5M]

Solution:

We have,



$$f(z) = r^5 \cos m\theta + i r^m \sin 5\theta$$

$$u = r^5 \cos m\theta$$

$$u_r = \cos m\theta [5r^4]$$

$$u_\theta = r^5 [-\sin m\theta \times m]$$

$$v = r^m \sin 5\theta$$

$$v_r = \sin 5\theta [mr^{m-1}]$$

$$v_\theta = r^m [\cos 5\theta \times 5]$$

Since, $f(z)$ is given analytic, we know that

$$ru_r = v_\theta$$

$$r \cos m\theta [5r^4] = r^m [\cos 5\theta \times 5]$$

$$5r^5 \cos m\theta = 5r^m \cos 5\theta$$

On Comparison, we get

$$\boxed{m = 5}$$

21. Find the value of k if $f(z) = r^3 \cos k\theta + i r^k \sin k\theta$ is analytic.

[M16/ElexExtcElectBiomInst/5M]

Solution:

We have,

$$f(z) = r^3 \cos k\theta + i r^k \sin k\theta$$

$$u = r^3 \cos k\theta$$

$$u_r = \cos k\theta [3r^2]$$

$$u_\theta = r^3 [-\sin k\theta \times k]$$

$$v = r^k \sin k\theta$$

$$v_r = \sin k\theta [kr^{k-1}]$$

$$v_\theta = r^k [\cos k\theta \times k]$$

Since, $f(z)$ is given analytic, we know that

$$ru_r = v_\theta$$

$$r \cos k\theta [3r^2] = r^k [\cos k\theta \times k]$$

$$3r^3 \cos k\theta = kr^k \cos k\theta$$

On Comparison, we get

$$\boxed{k = 3}$$

Type II: Harmonic Functions

1. Show that the following function is harmonic $e^x \cos y + x^3 - 3xy^2$
2. Check if the following function $f(r, \theta) = \left[r + \frac{a^2}{r}\right] \cos \theta$ is harmonic

[N18/M19/MTRX/5M]

Solution:

We have,

$$\text{Let } u = r \cos \theta + \frac{a^2 \cos \theta}{r}$$

$$u_r = \cos \theta [1] + a^2 \cos \theta \left[-\frac{1}{r^2}\right]$$

$$u_{rr} = 0 + a^2 \cos \theta \left[\frac{2}{r^3}\right]$$

$$u_\theta = r[-\sin \theta] + \frac{a^2}{r}[-\sin \theta]$$

$$u_{\theta\theta} = r[-\cos \theta] + \frac{a^2}{r}[-\cos \theta]$$

$$\text{By Laplace equation, } u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$$

$$\begin{aligned} u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} &= \frac{2a^2 \cos \theta}{r^3} + \frac{\cos \theta}{r} - \frac{a^2 \cos \theta}{r^3} - \frac{r \cos \theta}{r^2} - \frac{a^2 \cos \theta}{r^3} \\ &= \frac{2a^2 \cos \theta + r^2 \cos \theta - a^2 \cos \theta - r^2 \cos \theta - a^2 \cos \theta}{r^3} \\ &= 0 \end{aligned}$$

Hence, Laplace equation is verified and u is harmonic

3. Prove that real and imaginary parts of an analytic function $f(z) = u + iv$ are harmonic functions.

[N13/ElexExtcElectBiomInst/5M]

Solution:

We have the analytic function, $f(z) = u + iv$

Thus, we know that

$$u_x = v_y \quad \&$$

Diff w.r.t x , we get

$$u_{xx} = v_{yx}$$

Adding the two results, we get

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$$

$\therefore u$ is harmonic

Also,

$$v_x = -u_y \quad \&$$

Diff w.r.t x , we get

$$v_{xx} = -u_{yx}$$

Adding the two results, we get

$$v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0$$

$$u_y = -v_x$$

Diff w.r.t y , we get

$$u_{yy} = -v_{xy}$$

[considering $v_{yx} = v_{xy}$]

$$v_y = u_x$$

Diff w.r.t y , we get

$$v_{yy} = -u_{xy}$$

[considering $u_{yx} = u_{xy}$]



$\therefore v$ is harmonic.

Thus, the real and imaginary parts of an analytic function $f(z) = u + iv$ are harmonic functions

4. If $u(x, y)$ is harmonic function, then prove that $f(z) = u_x - iu_y$ is analytic function.

[M16/CompIT/5M]

Solution:

It is given that u is harmonic,

$$\text{i.e. } u_{xx} + u_{yy} = 0$$

$$u_{xx} = -u_{yy} \dots\dots\dots(1)$$

$$\text{Now, consider } f(z) = u_x - iu_y = U + iV$$

where,

$$U = u_x$$

&

$$V = -u_y$$

$$\therefore U_x = u_{xx}$$

$$\therefore V_x = -u_{yx}$$

$$\therefore U_y = u_{xy}$$

$$\therefore V_y = -u_{yy}$$

$$\text{From (1), } -u_{yy} = u_{xx}$$

$$\therefore V_y = u_{xx}$$

$$\text{Thus, we see that } U_x = V_y \text{ \& } U_y = -V_x$$

$$\therefore f(z) = u_x - iu_y \text{ is analytic}$$

5. If u and v are conjugate harmonic functions prove that uv is also harmonic

[M18/Comp/6M]

Solution:

Let $f(z) = u + iv$ be the analytic function.

$$\therefore u_x = v_y, u_y = -v_x$$

And u, v are harmonic

$$\therefore u_{xx} + u_{yy} = 0, v_{xx} + v_{yy} = 0 \dots\dots\dots(1)$$

$$\text{Now, } \frac{\partial}{\partial x}(uv) = u v_x + v u_x$$

$$\therefore \frac{\partial^2}{\partial x^2}(uv) = [u v_{xx} + v_x u_x] + [v u_{xx} + u_x v_x]$$

$$\therefore \frac{\partial^2}{\partial x^2}(uv) = uv_{xx} + vu_{xx} + 2u_x v_x \dots\dots\dots(2)$$

Similarly, we can get

$$\frac{\partial^2}{\partial y^2}(uv) = uv_{yy} + vu_{yy} + 2u_y v_y$$

$$\text{But } u_x = v_y \text{ and } u_y = -v_x$$

$$\therefore \frac{\partial^2}{\partial y^2}(uv) = uv_{yy} + vu_{yy} - 2u_x v_x \dots\dots\dots(3)$$



Adding (2) and (3),

$$\therefore \frac{\partial^2}{\partial x^2}(uv) + \frac{\partial^2}{\partial y^2}(uv) = u[v_{xx} + v_{yy}] + v[u_{xx} + u_{yy}]$$

From (1),

$$\therefore \frac{\partial^2}{\partial x^2}(uv) + \frac{\partial^2}{\partial y^2}(uv) = u[0] + v[0] = 0$$

$\therefore uv$ is harmonic.

6. If ϕ and ψ are functions of x and y satisfying Laplace equation then show that $s + it$ is holomorphic (analytic) where $s = \frac{\delta\phi}{\delta y} - \frac{\delta\psi}{\delta x}$ and $t = \frac{\delta\phi}{\delta x} + \frac{\delta\psi}{\delta y}$
7. Check whether the following function harmonic or not:

$$3x^2 + \sin x + y^2 + 5y + 4$$

[N19/IT/5M]

Solution:

$$\text{Let } u = 3x^2 + \sin x + y^2 + 5y + 4$$

$$u_x = 6x + \cos x$$

$$u_{xx} = 6 - \sin x$$

$$u_y = 2y + 5$$

$$u_{yy} = 2$$

Now,

$$u_{xx} + u_{yy} = 6 - \sin x + 2 \neq 0$$

$\therefore u$ is not harmonic

8. Determine the constant b such that $u = e^{bx} \cdot \cos 5y$ is harmonic.
Ans. ± 5
9. Find a and b such that $u = (5x + 3y)(2x^2 + axy + by^2)$ is harmonic
Ans. $-120, 66$

Type III: Finding an analytic function

1. Find the analytic function and its imaginary part if real part is

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

[N13/ElexExtcElectBiomInst/6M]

Solution:

We have,

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

$$u_x = [3x^2] - 3y^2[1] + [6x]$$

$$u_y = -3x[2y] - [6y]$$

We know that,

$$f'(z) = u_x + iv_x$$

$$\text{But } v_x = -u_y$$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [3z^2 + 6z] - i[0]$$

Integrating, we get

$$f(z) = 3 \left[\frac{z^3}{3} \right] + 6 \left[\frac{z^2}{2} \right] + C$$

$$f(z) = z^3 + 3z^2 + C$$

Now,

$$v_y = u_x = 3x^2 - 3y^2 + 6x$$

Integrating w.r.t. y , we get

$$v = 3x^2[y] - 3 \left[\frac{y^3}{3} \right] + 6x[y] + C_1 \dots \dots \dots (1)$$

Also,

$$v_x = -u_y = 6xy + 6y$$

Integrating w.r.t. x , we get

$$v = 6y \left[\frac{x^2}{2} \right] + 6y[x] + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$v = 3x^2y - y^3 + 6xy$$

2. Show that the function $u = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$ satisfies Laplace's equation and find its corresponding analytic function.

[N13/AutoMechCivil/5M][N14/ElexExtcElectBiomInst/8M]

Solution:

We have,

$$u = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$$

$$u_x = \cosh y [\cos x] + 2 \sinh y [-\sin x] + [2x] + 4y[1]$$

$$u_{xx} = \cosh y [-\sin x] + 2 \sinh y [-\cos x] + [2]$$



$$u_y = \sin x [\sin y] + 2 \cos x [\cos y] - [2y] + 4x[1]$$

$$u_{yy} = \sin x [\cos y] + 2 \cos x [\sin y] - [2]$$

Now,

$$u_{xx} + u_{yy} = -\sin x \cos y - 2 \cos x \sin y + 2 + \sin x \cos y + 2 \cos x \sin y - 2$$

$$\therefore u_{xx} + u_{yy} = 0$$

Thus, u satisfies Laplace equation

Now,

$$f'(z) = u_x + i v_x$$

$$\text{But } u_y = -v_x$$

$$f'(z) = u_x - i u_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - i u_{y(z,0)}$$

$$f'(z) = [\cos z + 2z] - i[2 \cos z + 4z]$$

$$f'(z) = (1 - 2i)[\cos z + 2z]$$

Integrating, we get

$$f(z) = (1 - 2i)\left[\sin z + 2 \cdot \frac{z^2}{2}\right] + C$$

$$f(z) = (1 - 2i)[\sin z + z^2] + C$$

3. Construct the analytic function whose real part is $e^x(x \cos y - y \sin y)$
[N14/CompIT/6M][M19/Elex/6M]

Solution:

We have,

$$u = e^x x \cos y - e^x y \sin y$$

$$u_x = \cos y [e^x(1) + x(e^x)] - y \sin y [e^x]$$

$$u_y = e^x x [-\sin y] - e^x [y(\cos y) + \sin y(1)]$$

Now,

$$f'(z) = u_x + i v_x$$

$$\text{But } v_x = -u_y$$

$$f'(z) = u_x - i u_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - i u_{y(z,0)}$$

$$f'(z) = [\cos 0(e^z + z e^z) - 0] - i[0]$$

$$f'(z) = e^z z + e^z$$

Integrating, we get

$$f(z) = [z(e^z) - (1)(e^z)] + e^z + C$$

$$f(z) = z e^z + C$$

4. Find the analytic function whose real part is
 $u = \frac{x}{2} \log(x^2 + y^2) - y \tan^{-1} \frac{y}{x} + \cos x \cosh y$

[M16/ElexExtcElectBiomInst/6M]

Solution:

We have,

$$u = \frac{x}{2} \log(x^2 + y^2) - y \tan^{-1} \frac{y}{x} + \cos x \cosh y$$

$$u_x = \left[\frac{x}{2} \left(\frac{1}{x^2 + y^2} \times 2x \right) + \log(x^2 + y^2) \left(\frac{1}{2} \right) \right] - y \left[\frac{1}{1 + \left(\frac{y}{x} \right)^2} \times -\frac{y}{x^2} \right] + \cosh y [-\sin x]$$

$$u_y = \frac{x}{2} \left[\frac{1}{x^2 + y^2} \times 2y \right] - \left[y \left(\frac{1}{1 + \left(\frac{y}{x} \right)^2} \times \frac{1}{x} \right) + \tan^{-1} \frac{y}{x} (1) \right] + \cos x [\sinh y]$$

Now,

$$f'(z) = u_x + i v_x$$

But $v_x = -u_y$

$$f'(z) = u_x - i u_y$$

By Milne – Thomson method,

$$f'(z) = u_{x(z,0)} - i u_{y(z,0)}$$

$$f'(z) = \left[\frac{z}{2} \left(\frac{2z}{z^2} \right) + \frac{1}{2} \log z^2 - \sin z \right] - i[0]$$

$$f'(z) = [1 + \log z - \sin z]$$

Integrating w.r.t z , we get,

$$f(z) = z + (z \log z - z) - (-\cos z) + c$$

$$f(z) = z \log z + \cos z + c$$

5. Find the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y + \cos 2x}$

[N15/AutoMechCivil/8M][M16/AutoMechCivil/6M][N17/IT/6M]

[M18/Elect/6M][M18/Inst/6M][M19/Biom/6M]

Solution:

We have,

$$u = \frac{\sin 2x}{\cosh 2y + \cos 2x}$$

$$u_x = \frac{(\cosh 2y + \cos 2x)[2 \cos 2x] - \sin 2x[-2 \sin 2x]}{(\cosh 2y + \cos 2x)^2}$$

$$u_y = \frac{(\cosh 2y + \cos 2x)[0] - \sin 2x[2 \sinh 2y]}{(\cosh 2y + \cos 2x)^2}$$

Now,

$$f'(z) = u_x + i v_x$$

But $v_x = -u_y$

$$f'(z) = u_x - i u_y$$

By Milne – Thomson method,

$$f'(z) = u_{x(z,0)} - i u_{y(z,0)}$$



$$f'(z) = \left[\frac{(1+\cos 2z)[2\cos 2z] - \sin 2z[-2\sin 2z]}{(1+\cos 2z)^2} \right] - i[0]$$

$$f'(z) = \left[\frac{2\cos 2z + 2\cos^2 2z + 2\sin^2 2z}{(1+\cos 2z)^2} \right]$$

$$f'(z) = \left[\frac{2\cos 2z + 2}{(1+\cos 2z)^2} \right]$$

$$f'(z) = \left[\frac{2(1+\cos 2z)}{(1+\cos 2z)^2} \right]$$

$$f'(z) = \left[\frac{2}{1+\cos 2z} \right]$$

$$f'(z) = \left[\frac{2}{2\cos^2 z} \right]$$

$$f'(z) = \sec^2 z$$

Integrating w.r.t z, we get,

$$f(z) = \tan z + c$$

6. Prove that there doesn't exist an analytic function whose real part is $u = e^x \sin y + x^2 + 3xy$
7. Verify Laplace's Equation for $u = \left[r + \frac{a^2}{r} \right] \cos \theta$. Also find v and $f(z)$.

[M14/CompIT/6M][N17/Biot/5M][N18/Biot/6M]

Solution:

We have,

$$u = r \cos \theta + \frac{a^2 \cos \theta}{r}$$

$$u_r = \cos \theta [1] + a^2 \cos \theta \left[-\frac{1}{r^2} \right]$$

$$u_{rr} = 0 + a^2 \cos \theta \left[\frac{2}{r^3} \right]$$

$$u_\theta = r[-\sin \theta] + \frac{a^2}{r}[-\sin \theta]$$

$$u_{\theta\theta} = r[-\cos \theta] + \frac{a^2}{r}[-\cos \theta]$$

By Laplace equation, $u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} = 0$

$$\begin{aligned} u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} &= \frac{2a^2 \cos \theta}{r^3} + \frac{\cos \theta}{r} - \frac{a^2 \cos \theta}{r^3} - \frac{r \cos \theta}{r^2} - \frac{a^2 \cos \theta}{r^3} \\ &= \frac{2a^2 \cos \theta + r^2 \cos \theta - a^2 \cos \theta - r^2 \cos \theta - a^2 \cos \theta}{r^3} \\ &= 0 \end{aligned}$$

Hence, Laplace equation is verified.

Now,

$$ru_r = v_\theta$$

$$v_\theta = r \cos \theta - \frac{a^2 \cos \theta}{r}$$

Integrating w.r.t. θ , we get

$$v = r[\sin \theta] - \frac{a^2}{r}[\sin \theta] + C_1 \dots \dots \dots (1)$$

Also,



$$rv_r = -u_\theta$$

$$v_r = -\frac{1}{r}u_\theta$$

$$v_r = \sin\theta + \frac{a^2 \sin\theta}{r^2}$$

Integrating w.r.t. r , we get

$$v = \sin\theta[r] + a^2 \sin\theta \left[-\frac{1}{r}\right] + C_2 \dots\dots\dots(2)$$

From (1) & (2), we get

$$v = r\sin\theta - \frac{a^2 \sin\theta}{r}$$

Now,

$$f(z) = u + iv$$

$$f(z) = \left[r\cos\theta + \frac{a^2 \cos\theta}{r}\right] + i \left[r\sin\theta - \frac{a^2 \sin\theta}{r}\right]$$

By Milne Thomson method,

$$f(z) = \left[z\cos 0 + \frac{a^2 \cos 0}{z}\right] + i \left[z\sin 0 - \frac{a^2 \sin 0}{z}\right]$$

$$f(z) = z + \frac{a^2}{z}$$

8. Find the analytic function $f(z)$ whose real part is $r^2 \cos 2\theta - r \cos \theta + 2$
[M14/ElexExtcElectBiomInst/6M]

Solution:

We have,

$$u = r^2 \cos 2\theta - r \cos \theta + 2$$

$$u_r = \cos 2\theta [2r] - \cos \theta [1]$$

$$u_\theta = r^2 [-2 \sin 2\theta] - r [-\sin \theta]$$

We know that,

$$f'(z) = e^{-i\theta} [u_r + iv_r]$$

$$\text{But } v_r = -\frac{1}{r}u_\theta$$

$$f'(z) = e^{-i\theta} \left[u_r - \frac{i}{r}u_\theta\right]$$

$$f'(z) = e^{-i\theta} \left[(2r \cos 2\theta - \cos \theta) - \frac{i}{r}(-2r^2 \sin 2\theta + r \sin \theta)\right]$$

By Milne Thomson method,

$$f'(z) = e^0 \left[(2z \cos 0 - \cos 0) - \frac{i}{z}(-2z^2 \sin 0 + z \sin 0)\right]$$

$$f'(z) = 2z - 1$$

Integrating, we get

$$f(z) = z^2 - z + C$$

9. Find the constant 'a' so that the function $u = x^2 + ay^2$ is harmonic. Also find harmonic conjugate.
Ans. $a = -1, v = 2xy$

10. Find the analytic function whose real part is $e^{-x}(y \cos y - x \sin y)$
[N15/ChemBiot/6M][N17/Chem/6M][M18/Biot/6M][N18/Chem/6M]

Solution:

We have,

$$u = e^{-x}y \cos y - e^{-x}x \sin y$$

$$u_x = y \cos y [-e^{-x}] - \sin y [e^{-x}(1) + x(-e^{-x})]$$

$$u_y = e^{-x}[y(-\sin y) + \cos y(1)] - e^{-x}x[\cos y]$$

Now,

$$f'(z) = u_x + iv_x$$

$$\text{But } v_x = -u_y$$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [0] - i[e^{-z}[0 + \cos 0] - e^{-z}z[\cos 0]]$$

$$f'(z) = -i[e^{-z} - e^{-z}z]$$

Integrating, we get

$$f(z) = -i \left[\frac{e^{-z}}{-1} - z \left(\frac{e^{-z}}{-1} \right) + (1) \left(\frac{e^{-z}}{1} \right) \right] + C$$

$$f(z) = -ize^{-z} + C$$

11. Construct an analytic function whose real part is $e^{2x}(x \cos 2y - y \sin 2y)$
[M16/ChemBiot/6M][N17/Extc/6M]

Solution:

$$u = e^{2x}(x \cos 2y - y \sin 2y)$$

$$u_x = \cos 2y [e^{2x}(1) + x(2e^{2x})] - y \sin 2y [2e^{2x}]$$

$$u_y = e^{2x}x[-2 \sin 2y] - e^{2x}[y(2 \cos 2y) + \sin 2y(1)]$$

We know that,

$$f'(z) = u_x + iv_x$$

$$\text{But } v_x = -u_y$$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [e^{2z} + 2ze^{2z}] - i[0]$$

Integrating, we get

$$f(z) = \frac{e^{2z}}{2} + 2z \left(\frac{e^{2z}}{2} \right) - 2 \left(\frac{e^{2z}}{4} \right) + C$$

$$f(z) = ze^{2z} + C$$

12. Find the imaginary part of the analytic function whose real part is $e^{2x}(x\cos 2y - y\sin 2y)$. Also verify that v is harmonic

[N13/Chem/7M]

Solution:

We have,

$$u = e^{2x}(x\cos 2y - y\sin 2y)$$

$$u_x = \cos 2y[e^{2x}(1) + x(2e^{2x})] - y\sin 2y[2e^{2x}]$$

$$u_y = e^{2x}x[-2\sin 2y] - e^{2x}[y(2\cos 2y) + \sin 2y(1)]$$

We know that,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [e^{2z} + 2ze^{2z}] - i[0]$$

Integrating, we get

$$f(z) = \frac{e^{2z}}{2} + 2z\left(\frac{e^{2z}}{2}\right) - 2\left(\frac{e^{2z}}{4}\right) + C$$

$$f(z) = ze^{2z} + C$$

$$f(z) = (x + iy)e^{2(x+iy)} + C$$

$$f(z) = (x + iy)e^{2x}e^{2iy} + C$$

$$f(z) = e^{2x}(x + iy)(\cos 2y + i\sin 2y) + C$$

$$f(z) = e^{2x}[x\cos 2y + ix\sin 2y + iycos 2y + i^2y\sin 2y] + C$$

$$f(z) = e^{2x}(x\cos 2y - y\sin 2y) + ie^{2x}(x\sin 2y + y\cos 2y) + C$$

$$\therefore v = e^{2x}(x\sin 2y + y\cos 2y)$$

$$v_x = \sin 2y[e^{2x}(1) + x(2e^{2x})] + y\cos 2y[2e^{2x}]$$

$$v_x = 2e^{2x}x\sin 2y + 2e^{2x}y\cos 2y + e^{2x}\sin 2y$$

$$v_{xx} = \sin 2y[2e^{2x}(1) + 2x(2e^{2x})] + 2y\cos 2y(2e^{2x}) + \sin 2y(2e^{2x})$$

$$v_{xx} = 2e^{2x}[2x\sin 2y + 2y\cos 2y + 2\sin 2y] \dots\dots\dots(1)$$

$$v_y = e^{2x}x[2\cos 2y] + e^{2x}[y(-2\sin 2y) + \cos 2y(1)]$$

$$v_y = e^{2x}\cos 2y + 2xe^{2x}\cos 2y - 2e^{2x}y\sin 2y$$

$$v_{yy} = e^{2x}[-2\sin 2y] + 2xe^{2x}[-2\sin 2y] - 2e^{2x}[y(2\cos 2y) + \sin 2y(1)]$$

$$v_{yy} = -2e^{2x}[2x\sin 2y + 2y\cos 2y + 2\sin 2y] \dots\dots\dots(2)$$

Adding (1) & (2), we get

$$v_{xx} + v_{yy} = 0$$

$\therefore v$ is harmonic

13. Show that the function $u = 2x(1 - y)$ is a harmonic function. Find its harmonic conjugate and corresponding analytic function

[M19/Comp/8M]



Solution:

We have,

$$u = 2x - 2xy$$

$$u_x = 2[1] - 2y[1]$$

$$u_{xx} = 0$$

$$u_y = 0 - 2x[1]$$

$$u_{yy} = 0$$

Now,

$$u_{xx} + u_{yy} = 0 + 0 = 0$$

Thus, u is a harmonic function

Now,

$$v_y = u_x = 2 - 2y$$

Integrating w.r.t. y , we get

$$v = 2[y] - 2\left[\frac{y^2}{2}\right] + C_1 \dots \dots \dots (1)$$

Also,

$$v_x = -u_y = 2x$$

Integrating w.r.t. x , we get

$$v = 2\left[\frac{x^2}{2}\right] + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$v = 2y - y^2 + x^2$$

Now,

$$f(z) = u + iv$$

By Milne Thomson method,

$$f(z) = u_{(z,0)} + iv_{(z,0)}$$

$$f(z) = [2z - 2z(0)] + i[0 - 0 + z^2]$$

$$f(z) = 2z + iz^2$$

14. Prove that $u = x^2 - y^2$ is harmonic also find corresponding analytic function $f(z)$

[N19/Elex/5M]

Solution:

We have,

$$u = x^2 - y^2$$

$$u_x = 2x$$

$$u_{xx} = 2$$

$$u_y = 0 - 2y$$

$$u_{yy} = -2$$

Now,

$$u_{xx} + u_{yy} = 2 - 2 = 0$$



Thus, u is a harmonic function

Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [2z] - i[0]$$

Integrating, we get

$$f(z) = \frac{2z^2}{2} + C$$

$$f(z) = z^2 + C$$

15. Show that the given function is harmonic and find the corresponding analytic function

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

[N18/Elex/5M]

Solution:

We have,

$$u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$$

$$u_x = [3x^2] - 3y^2[1] + [6x]$$

$$u_{xx} = 6x - 0 + 6$$

$$u_y = -3x[2y] - [6y]$$

$$u_{yy} = -6x - 6$$

We see that, $u_{xx} + u_{yy} = 6x + 6 - 6x - 6 = 0$

Thus, u is harmonic

We know that,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [3z^2 + 6z] - i[0]$$

Integrating, we get

$$f(z) = 3 \left[\frac{z^3}{3} \right] + 6 \left[\frac{z^2}{2} \right] + C$$

$$f(z) = z^3 + 3z^2 + C$$

16. Construct an analytic function whose real part is

$$e^{-x}\{(x^2 - y^2)\cos y + 2xysiny)\}$$

[N17/AutoMechCivil/6M][M18/AutoMechCivil/6M]

[N18/AutoMechCivil/6M][N19/AutoMechCivil/6M][M19/MTRX/6M]

Solution:

We have,

$$u = e^{-x}\{(x^2 - y^2)\cos y + 2xysiny)\}$$

$$u_x = \cos y[e^{-x}(2x) + (x^2 - y^2)(-e^{-x})] + 2ysiny[e^{-x}(1) + x(-e^{-x})]$$

$$u_y = e^{-x}[(x^2 - y^2)(-siny) + \cos y(-2y)] + 2xe^{-x}[y(\cos y) + siny(1)]$$

Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne – Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [2ze^{-z} - z^2e^{-z}] - i[0]$$

$$f'(z) = (2z - z^2)e^{-z}$$

Integrating w.r.t z, we get,

$$f(z) = (2z - z^2) \left(\frac{e^{-z}}{-1} \right) - (2 - 2z) \left(\frac{e^{-z}}{1} \right) + (-2) \left(\frac{e^{-z}}{-1} \right) + c$$

$$f(z) = e^{-z}(-2z + z^2 - 2 + 2z + 2) + c$$

$$f(z) = z^2e^{-z} + c$$

17. Construct an analytic function whose real part is $e^x \cos y$

[M19/Extc/6M]

Solution:

We have,

$$u = e^x \cos y$$

$$u_x = \cos y[e^x]$$

$$u_y = e^x[-siny]$$

Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne – Thomson method,

$$f'(z) = u_{x(z,0)} - iu_{y(z,0)}$$

$$f'(z) = [\cos 0(e^z)] - i[e^z(-\sin 0)]$$

$$f'(z) = e^z$$

Integrating w.r.t z, we get,

$$f(z) = e^z + c$$



18. Prove that $u = e^{2x} \cos 2y$ is harmonic function. Hence find its corresponding harmonic conjugate function

[M18/Elect/6M]

Solution:

We have,

$$u = e^{2x} \cos 2y$$

$$u_x = \cos 2y [2e^{2x}]$$

$$u_{xx} = \cos 2y [4e^{2x}]$$

$$u_y = e^{2x} [-2 \sin 2y]$$

$$u_{yy} = e^{2x} [-4 \sin 2y]$$

Now,

$$u_{xx} + u_{yy} = 4e^{2x} \cos 2y - 4e^{2x} \cos 2y = 0$$

Thus, u is harmonic function

Now,

$$v_y = u_x = 2e^{2x} \cos 2y$$

Integrating w.r.t. y , we get

$$v = 2e^{2x} \left[\frac{\sin 2y}{2} \right] + C_1 \dots \dots \dots (1)$$

Also,

$$v_x = -u_y = 2e^{2x} \sin 2y$$

Integrating w.r.t. x , we get

$$v = 2 \sin 2y \left[\frac{e^{2x}}{2} \right] + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$v = e^{2x} \sin 2y$$

19. Show that there does not exist an analytic function whose real part is $3x^2 - 2x^2y + y^2$

[N17/IT/5M]

Solution:

$$\text{Let } u = 3x^2 - 2x^2y + y^2$$

$$u_x = 6x - 4xy$$

$$u_{xx} = 6 - 4y$$

$$u_y = -2x^2 + 2y$$

$$u_{yy} = 2$$

We see that,

$$u_{xx} + u_{yy} = 6 - 4y + 2 \neq 0$$

$\therefore u$ is not harmonic

Thus, the given function cannot be a real part of an analytic function.

Therefore, there does not exist an analytic function whose real part is $3x^2 - 2x^2y + y^2$



20. State true or false with proper justification: 'There does not exist an analytic function whose real part is $x^3 - 3x^2y - y^3$.' Justify your answer.

[M14/CompIT/5M][N17/MTRX/5M]

Solution:

$$\text{Let } u = x^3 - 3x^2y - y^3$$

$$u_x = 3x^2 - 6xy$$

$$u_{xx} = 6x - 6y$$

$$u_y = -3x^2 - 3y^2$$

$$u_{yy} = -6y$$

We see that,

$$u_{xx} + u_{yy} = 6x - 6y - 6y \neq 0$$

$\therefore u$ is not harmonic

Thus, the given function cannot be a real part of an analytic function.

Therefore, there does not exist an analytic function whose real part is $x^3 - 3x^2y - y^3$

21. Show that $u = \cos x \cosh y$ is a harmonic function. Find its harmonic conjugate and the corresponding analytic function.

[N17/Comp/8M]

Solution:

$$u = \cos x \cosh y$$

$$u_x = -\sin x \cosh y$$

$$u_{xx} = -\cos x \cosh y$$

$$u_y = \cos x \sinh y$$

$$u_{yy} = \cos x \cosh y$$

Now,

$$u_{xx} + u_{yy} = -\cos x \cosh y + \cos x \cosh y = 0$$

Thus, u is a harmonic function

Now,

$$v_y = u_x = -\sin x \cosh y$$

Integrating w.r.t. y , we get

$$v = -\sin x [\sinh y] + C_1 \dots \dots \dots (1)$$

Also,

$$v_x = -u_y = -\cos x \sinh y$$

Integrating w.r.t. x , we get

$$v = -\sinh y [\sin x] + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$v = -\sin x \sinh y$$

Now,

$$f(z) = u + iv$$



By Milne Thomson method,

$$f(z) = u_{(z,0)} + iv_{(z,0)}$$

$$f(z) = [\cos z \cosh 0] + i[-\sin z \sinh 0]$$

$$f(z) = \cos z$$

22. Show that the function is harmonic and find the harmonic conjugate

$$u = \cos x \cosh y - 2xy$$

[N19/Comp/6M]

Solution:

$$u = \cos x \cosh y - 2xy$$

$$u_x = -\sin x \cosh y - 2y$$

$$u_{xx} = -\cos x \cosh y$$

$$u_y = \cos x \sinh y - 2x$$

$$u_{yy} = \cos x \cosh y - 0$$

Now,

$$u_{xx} + u_{yy} = -\cos x \cosh y + \cos x \cosh y = 0$$

Thus, u is a harmonic function

Now,

$$v_y = u_x = -\sin x \cosh y - 2y$$

Integrating w.r.t. y , we get

$$v = -\sin x [\sinh y] - y^2 + C_1 \dots \dots \dots (1)$$

Also,

$$v_x = -u_y = -\cos x \sinh y + 2x$$

Integrating w.r.t. x , we get

$$v = -\sinh y [\sin x] + x^2 + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$v = -\sin x \sinh y + x^2 - y^2$$

23. Show that $u = y^3 - 3x^2y$ is a harmonic function. Find its harmonic conjugate and the corresponding analytic function.

[M15/ChemBiot/6M]

Solution:

We have,

$$u = y^3 - 3x^2y$$

$$u_x = -6xy$$

$$u_{xx} = -6y$$

$$u_y = 3y^2 - 3x^2$$

$$u_{yy} = 6y$$

Now,

$$u_{xx} + u_{yy} = -6y + 6y = 0$$



Thus, u is a harmonic function

Now,

$$v_y = u_x = -6xy$$

Integrating w.r.t. y , we get

$$v = -6x \left[\frac{y^2}{2} \right] + C_1 \dots \dots \dots (1)$$

Also,

$$v_x = -u_y = -3y^2 + 3x^2$$

Integrating w.r.t. x , we get

$$v = -3y^2 \left[x \right] + 3 \left[\frac{x^3}{3} \right] + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$v = x^3 - 3xy^2$$

Now,

$$f(z) = u + iv$$

By Milne Thomson method,

$$f(z) = u_{(z,0)} + iv_{(z,0)}$$

$$f(z) = [0 - 3z^2(0)] + i[z^3 - 3z(0)]$$

$$f(z) = iz^3$$

24. Show that $u = y^3 - 3x^2y$ is a harmonic function. Find its harmonic conjugate

[M17/ElexExtcElectBiomInst/5M]

Solution:

$$u = y^3 - 3x^2y$$

$$u_x = -6xy$$

$$u_{xx} = -6y$$

$$u_y = 3y^2 - 3x^2$$

$$u_{yy} = 6y$$

Now,

$$u_{xx} + u_{yy} = -6y + 6y = 0$$

Thus, u is a harmonic function

We know that,

$$u_x = v_y$$

$$v_y = -6xy$$

Integrating w.r.t y , we get

$$v = -6x \left[\frac{y^2}{2} \right] + c_1 \dots \dots \dots (1)$$

Also,

$$u_y = -v_x$$

$$v_x = -u_y$$



$$v_x = -3y^2 + 3x^2$$

Integrating w.r.t x, we get

$$v = -3y^2[x] + 3\left[\frac{x^3}{3}\right] + c_2 \dots\dots\dots(2)$$

From (1) & (2), we get

$$v = x^3 - 3xy^2$$

25. Show that $u = x^4 - 6x^2y^2 + y^4$ is harmonic function. Find its harmonic conjugate and corresponding analytic function. Ans. $z^4 + C$

26. Show that the function $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic. Find its harmonic conjugate and corresponding analytic function

[N19/Extc/6M]

Solution:

We have,

$$u = 3x^2y + 2x^2 - y^3 - 2y^2$$

$$u_x = 6xy + 4x$$

$$u_{xx} = 6y + 4$$

$$u_y = 3x^2 - 3y^2 - 4y$$

$$u_{yy} = -6y - 4$$

Now,

$$u_{xx} + u_{yy} = 6y + 4 - 6y - 4 = 0$$

Thus, u is a harmonic function

Now,

$$v_y = u_x = 6xy + 4x$$

Integrating w.r.t. y, we get

$$v = 6x\left[\frac{y^2}{2}\right] + 4x[y] + C_1 \dots\dots\dots(1)$$

Also,

$$v_x = -u_y = -3x^2 + 3y^2 + 4y$$

Integrating w.r.t. x, we get

$$v = -3\left[\frac{x^3}{3}\right] + 3y^2[x] + 4y[x] + C_2 \dots\dots\dots(2)$$

From (1) & (2),

$$v = 3xy^2 + 4xy - x^3$$

Now,

$$f(z) = u + iv$$

By Milne Thomson method,

$$f(z) = u_{(z,0)} + iv_{(z,0)}$$

$$f(z) = [0 + 2z^2 - 0 - 0] + i[0 + 0 - z^3]$$

$$f(z) = 2z^2 - iz^3$$

27. Show that $u = \frac{1}{2} \log(x^2 + y^2)$ satisfies Laplace's equation and find its corresponding analytic function and the harmonic conjugate.

Ans. $\log z + c$

28. Find the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y - \cos 2x}$

Ans. $\cot z + c$

29. Determine the analytic function $f(z)$ where $u = -r^3 \sin 3\theta$
[N16/AutoMechCivil/5M][N17/Elex/6M]

Solution:

We have,

$$u = -r^3 \sin 3\theta$$

$$u_r = \sin 3\theta [-3r^2]$$

$$u_\theta = -r^3 [3 \cos 3\theta]$$

We know that,

$$f'(z) = e^{-i\theta} [u_r + i v_r]$$

$$\text{But } v_r = -\frac{1}{r} u_\theta$$

$$f'(z) = e^{-i\theta} \left[u_r - \frac{i}{r} u_\theta \right]$$

$$f'(z) = e^{-i\theta} \left[(-3r^2 \sin 3\theta) - \frac{i}{r} (-3r^3 \cos 3\theta) \right]$$

By Milne Thomson method,

$$f'(z) = e^0 \left[(0) - \frac{i}{z} (-3z^3 \cos 0) \right]$$

$$f'(z) = 3iz^2$$

Integrating, we get

$$f(z) = iz^3 + c$$

30. Prove that $u = -r^3 \sin 3\theta$ is harmonic function also find harmonic conjugate function of u

[M18/Elex/5M]

Solution:

$$u = -r^3 \sin 3\theta$$

$$u_r = \sin 3\theta [-3r^2]$$

$$u_{rr} = \sin 3\theta [-6r]$$

$$u_\theta = -r^3 [3 \cos 3\theta]$$

$$u_{\theta\theta} = -r^3 [-9 \sin 3\theta]$$

Now, consider

$$\begin{aligned} u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} &= (-6r \sin 3\theta) + \frac{1}{r} (-3r^2 \sin 3\theta) + \frac{1}{r^2} (9r^3 \sin 3\theta) \\ &= -6r \sin 3\theta - 3r \sin 3\theta + 9r \sin 3\theta \\ &= 0 \end{aligned}$$



Thus, u is harmonic

Now,

$$ru_r = v_\theta$$

$$\therefore v_\theta = -3r^3 \sin 3\theta$$

Integrating w.r.t θ , we get

$$v = -3r^3 \left[-\frac{\cos 3\theta}{3} \right] + C_1 \dots\dots\dots(1)$$

Also,

$$rv_r = -u_\theta$$

$$v_r = -\frac{1}{r}u_\theta$$

$$v_r = 3r^2 \cos 3\theta$$

Integrating w.r.t r , we get

$$v = 3\cos 3\theta \left[\frac{r^3}{3} \right] + C_2 \dots\dots\dots(2)$$

From (1) & (2),

$$v = r^3 \cos 3\theta$$

31. Does there exist an analytic function whose real part is $u = k(1 + \cos \theta)$?
Give justification.

[N17/AutoMechCivil/5M][N18/AutoMechCivil/5M]

Solution:

$$\text{Let } u = k(1 + \cos \theta)$$

$$u_r = 0$$

$$u_{rr} = 0$$

$$u_\theta = k(-\sin \theta)$$

$$u_{\theta\theta} = k(-\cos \theta)$$

We see that,

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0 + \frac{1}{r}(0) + \frac{1}{r^2}(-k\cos \theta) \neq 0$$

$\therefore u$ is not harmonic

Thus, the given function cannot be a real part of an analytic function.

Therefore, there does not exist an analytic function whose real part is $u = k(1 + \cos \theta)$

32. Show that $u = \left[r - \frac{a^2}{r^2} \right] \sin \theta$ cannot be real part of an analytic function

33. Show that $v = e^x \sin y$ is harmonic function. Find its harmonic conjugate and corresponding analytic function.

[N14/ChemBiot/7M][M15/ElexExtcElectBiomInst/6M]

[M16/CompIT/6M][M18/AutoMechCivil/6M][N18/Biot/5M]

Solution:



We have,

$$v = e^x \sin y$$

$$v_x = \sin y [e^x]$$

$$v_{xx} = \sin y [e^x]$$

$$v_y = e^x [\cos y]$$

$$v_{yy} = e^x [-\sin y]$$

Now,

$$v_{xx} + v_{yy} = e^x \sin y - e^x \sin y = 0$$

Thus, v is harmonic function

Now,

$$u_x = v_y = e^x \cos y$$

Integrating w.r.t. x , we get

$$u = [e^x] \cos y + C_1 \dots \dots \dots (1)$$

Also,

$$u_y = -v_x = -\sin y e^x$$

Integrating w.r.t. y , we get

$$u = -[-\cos y] e^x + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$u = e^x \cos y$$

Now,

$$f(z) = u + iv$$

By Milne Thomson method,

$$f(z) = u_{(z,0)} + iv_{(z,0)}$$

$$f(z) = [e^z \cos 0] + i[\sin 0 e^z]$$

$$f(z) = e^z$$

34. Show that the function is harmonic and find the harmonic conjugate

$$v = e^x \cos y + x^3 - 3xy^2$$

[N18/Comp/6M]

Solution:

We have,

$$v = e^x \cos y + x^3 - 3xy^2$$

$$v_x = \cos y [e^x] + 3x^2 - 3y^2 [1]$$

$$v_{xx} = \cos y [e^x] + 6x$$

$$v_y = e^x [-\sin y] - 3x [2y]$$

$$v_{yy} = -e^x \cos y - 6x$$

Now,

$$v_{xx} + v_{yy} = e^x \cos y + 6x - e^x \cos y - 6x = 0$$

Thus, v is harmonic function

Now,



$$u_x = v_y = -e^x \sin y - 6xy$$

Integrating w.r.t. x , we get

$$u = -\sin y [e^x] - 6y \left[\frac{x^2}{2} \right] + C_1 \dots \dots \dots (1)$$

Also,

$$u_y = -v_x = -e^x \cos y - 3x^2 + 3y^2$$

Integrating w.r.t. y , we get

$$u = -e^x [\sin y] - 3x^2 [y] + 3 \left[\frac{y^3}{3} \right] + C_2 \dots \dots \dots (2)$$

From (1) & (2),

$$u = -e^x \sin y - 3x^2 y + y^3$$

35. Find the analytic function whose imaginary part is $\frac{\sinh 2y}{\cos 2x + \cosh 2y}$

[N17/Inst/6M]

Solution:

We have,

$$v = \frac{\sinh 2y}{\cosh 2y + \cos 2x}$$

$$v_x = \frac{(\cosh 2y + \cos 2x)[0] - \sinh 2y[-2 \sin 2x]}{(\cosh 2y + \cos 2x)^2}$$

$$v_y = \frac{(\cosh 2y + \cos 2x)[2 \cosh 2y] - \sin 2x[2 \sinh 2y]}{(\cosh 2y + \cos 2x)^2}$$

Now,

$$f'(z) = u_x + i v_x$$

But $u_x = v_y$

$$f'(z) = v_y + i v_x$$

By Milne – Thomson method,

$$f'(z) = v_{y(z,0)} + i v_{x(z,0)}$$

$$f'(z) = \left[\frac{(1 + \cos 2z)[2] - \sin 2z[0]}{(1 + \cos 2z)^2} \right] - i[0]$$

$$f'(z) = \left[\frac{2(1 + \cos 2z)}{(1 + \cos 2z)^2} \right]$$

$$f'(z) = \left[\frac{2}{1 + \cos 2z} \right]$$

$$f'(z) = \left[\frac{2}{2 \cos^2 z} \right]$$

$$f'(z) = \sec^2 z$$

Integrating w.r.t z , we get,

$$f(z) = \tan z + c$$

36. Find the analytic function whose imaginary part is

$$e^{-x} [2xy \cos y + (y^2 - x^2) \sin y] \quad \text{Ans. } e^{-z} \cdot z^2 + c$$

37. Prove that there does not exist an analytic function whose imaginary part is $3x^2 + \sin x + y^2 + 5y + 4$

[M19/IT/5M]

Solution:

$$\text{Let } v = 3x^2 + \sin x + y^2 + 5y + 4$$

$$v_x = 6x + \cos x$$

$$v_{xx} = 6 - \sin x$$

$$v_y = 2y + 5$$

$$v_{yy} = 2$$

We see that,

$$v_{xx} + v_{yy} = 6 - \sin x + 2 \neq 0$$

$\therefore v$ is not harmonic

Thus, the given function cannot be a real part of an analytic function.

Therefore, there does not exist an analytic function whose real part is $3x^2 + \sin x + y^2 + 5y + 4$

38. Find the analytic function whose imaginary part is $\cos x \cosh y + \frac{x}{x^2 + y^2}$

[N13/Biot/6M][N14/ElexExtcElectBiomInst/6M]

Solution:

We have,

$$v = \cos x \cosh y + \frac{x}{x^2 + y^2}$$

$$v_x = \cosh y [-\sin x] + \frac{(x^2 + y^2)[1] - x[2x]}{(x^2 + y^2)^2}$$

$$v_y = \cos x [\sinh y] + \frac{(x^2 + y^2)[0] - x[2y]}{(x^2 + y^2)^2}$$

Now,

$$f'(z) = u_x + i v_x$$

But $u_x = v_y$

$$f'(z) = v_y + i v_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + i v_{x(z,0)}$$

$$f'(z) = [0] + i \left[\cosh 0 (-\sin z) + \frac{z^2 - 2z^2}{(z^2)^2} \right]$$

$$f'(z) = i \left[-\sin z - \frac{1}{z^2} \right]$$

Integrating, we get

$$f(z) = i \left[\cos z + \frac{1}{z} \right] + C$$

39. Find the analytic function $f(z) = u + iv$ if $v = \frac{x}{x^2+y^2} + \cosh x \cos y$

[M17/CompIT/6M][N18/IT/6M]

Solution:

We have,

$$v = \cosh x \cos y + \frac{x}{x^2+y^2}$$

$$v_x = \cos y [\sinh x] + \frac{(x^2+y^2)[1]-x[2x]}{(x^2+y^2)^2}$$

$$v_y = \cosh x [-\sin y] + \frac{(x^2+y^2)[0]-x[2y]}{(x^2+y^2)^2}$$

Now,

$$f'(z) = u_x + iv_x$$

But $u_x = v_y$

$$f'(z) = v_y + iv_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + iv_{x(z,0)}$$

$$f'(z) = [0] + i \left[\cos 0 (\sinh z) + \frac{z^2 - 2z^2}{(z^2)^2} \right]$$

$$f'(z) = i \left[\sinh z - \frac{1}{z^2} \right]$$

Integrating, we get

$$f(z) = i \left[\cosh z + \frac{1}{z} \right] + C$$

40. If $v = 3x^2y + 6xy - y^3$ show that v is harmonic and find the corresponding analytic function.

[M14/AutoMechCivil/5M][N16/CompIT/6M][M18/MTRX/8M]

[M19/AutoMechCivil/6M]

Solution:

We have,

$$v = 3x^2y + 6xy - y^3$$

$$v_x = 3y[2x] + 6y[1]$$

$$v_{xx} = 3y[2]$$

$$v_y = 3x^2[1] + 6x[1] - [3y^2]$$

$$v_{yy} = -[6y]$$

Now,

$$v_{xx} + v_{yy} = 6y - 6y = 0$$

Thus, v is harmonic function.

Now,

$$f'(z) = u_x + iv_x$$

But $u_x = v_y$

$$f'(z) = v_y + iv_x$$



By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + i v_{x(z,0)}$$

$$f'(z) = [3z^2 + 6z] + i[0]$$

$$f'(z) = 3z^2 + 6z$$

Integrating, we get

$$f(z) = 3 \frac{z^3}{3} + 6 \frac{z^2}{2} + C$$

$$f(z) = z^3 + 3z^2 + C$$

41. Find the analytic function $f(z) = u + iv$ where $v = \tan^{-1} \frac{y}{x}$

[N19/Chem/6M]

Solution:

We have,

$$v = \tan^{-1} \frac{y}{x}$$

$$v_x = \frac{1}{1+\frac{y^2}{x^2}} \left(y \cdot \frac{-1}{x^2} \right) = \frac{-y}{x^2+y^2}$$

$$v_y = \frac{1}{1+\frac{y^2}{x^2}} \left(\frac{1}{x} \right) = \frac{x}{x^2+y^2}$$

Now,

$$f'(z) = u_x + i v_x$$

$$\text{But } u_x = v_y$$

$$f'(z) = v_y + i v_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + i v_{x(z,0)}$$

$$f'(z) = \left[\frac{z}{z^2+0} \right] + i[0]$$

$$f'(z) = \frac{1}{z}$$

Integrating, we get

$$f(z) = \log z + C$$

42. Find the analytic function whose imaginary part is

$$v = \log(x^2 + y^2) + x - 2y$$

[N18/Comp/6M]

Solution:

We have,

$$v = \log(x^2 + y^2) + x - 2y$$

$$v_x = \frac{1}{x^2+y^2} (2x) + 1$$

$$v_y = \frac{1}{x^2+y^2} (2y) - 2$$

Now,



$$f'(z) = u_x + iv_x$$

But $u_x = v_y$

$$f'(z) = v_y + iv_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + iv_{x(z,0)}$$

$$f'(z) = [0 - 2] + i \left[\frac{2z}{z^2} + 1 \right]$$

$$f'(z) = -2 + \frac{2i}{z} + i$$

Integrating, we get

$$f(z) = -2z + 2i \log z + iz + C$$

43. Show that $v = e^{2x}(y \cos 2y + x \sin 2y)$ is harmonic. Find the harmonic conjugate function u and the analytic function $f(z)$.

Ans. $ze^{2z} + c$

44. Find the analytic function $f(z) = u + iv$, if $v = e^x(x \sin y + y \cos y)$

[M18/IT/8M][M18/N18/Biom/6M]

Solution:

We have,

$$v = e^x x \sin y + e^x y \cos y$$

$$v_x = \sin y [e^x(1) + x(e^x)] + y \cos y [e^x]$$

$$v_y = e^x x [\cos y] + e^x [y(-\sin y) + \cos y(1)]$$

Now,

$$f'(z) = u_x + iv_x$$

But $u_x = v_y$

$$f'(z) = v_y + iv_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + iv_{x(z,0)}$$

$$f'(z) = [e^z z [\cos 0] + e^z [0 + \cos 0]] + i[0]$$

$$f'(z) = e^z z + e^z$$

Integrating, we get

$$f(z) = [z(e^z) - (1)(e^z)] + e^z + C$$

$$f(z) = ze^z + C$$

45. Find the analytic function whose imaginary part is $e^{-x}(y \sin y + x \cos y)$

[M17/ElexExtcElectBiomInst/6M]

Solution:

$$v = e^{-x} y \sin y + e^{-x} x \cos y$$

$$v_x = y \sin y [-e^{-x}] + \cos y [e^{-x}(1) + x(-e^{-x})]$$

$$v_y = e^{-x} [y(\cos y) + \sin y(1)] + e^{-x} x [-\sin y]$$



Now,

$$f'(z) = u_x + iv_x$$

But $u_x = v_y$

$$f'(z) = v_y + iv_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + iv_{x(z,0)}$$

$$f'(z) = [0] + i[0 + \cos 0][e^{-z} - ze^{-z}]$$

$$f'(z) = i[e^{-z} - e^{-z}z]$$

Integrating, we get

$$f(z) = i \left[\frac{e^{-z}}{-1} - z \left(\frac{e^{-z}}{-1} \right) + (1) \left(\frac{e^{-z}}{1} \right) \right] + C$$

$$f(z) = ize^{-z} + C$$

46. Show that $v = e^x(x \sin y + y \cos y)$ satisfies laplace equation and find its corresponding analytic function and its harmonic conjugate.

[N13/ElexExtcElectBiomInst/8M][M18/Comp/8M]

Solution:

We have,

$$v = e^x x \sin y + e^x y \cos y$$

$$v_x = \sin y [e^x (1) + x(e^x)] + y \cos y [e^x]$$

$$v_{xx} = \sin y [e^x + x(e^x) + e^x (1)] + y \cos y [e^x]$$

$$v_y = e^x x [\cos y] + e^x [y(-\sin y) + \cos y (1)]$$

$$v_{yy} = e^x x [-\sin y] + e^x [y(-\cos y) - \sin y (1) - \sin y]$$

Now,

$$v_{xx} + v_{yy} = 2e^x \sin y + xe^x \sin y + e^x y \cos y - xe^x \sin y - 2e^x \sin y - e^x y \cos y$$

$$\therefore v_{xx} + v_{yy} = 0$$

Thus, v satisfies Laplace equation

Now,

$$f'(z) = u_x + iv_x$$

But $u_x = v_y$

$$f'(z) = v_y + iv_x$$

By Milne Thomson method,

$$f'(z) = v_{y(z,0)} + iv_{x(z,0)}$$

$$f'(z) = [e^z z [\cos 0] + e^z [0 + \cos 0]] + i[0]$$

$$f'(z) = e^z z + e^z$$

Integrating, we get

$$f(z) = [z(e^z) - (1)(e^z)] + e^z + C$$

$$f(z) = ze^z + C$$

Now,



$$\begin{aligned}
 f(z) &= (x + iy)e^{x+iy} + C \\
 &= (x + iy)e^x e^{iy} + C \\
 &= e^x(x + iy)(\cos y + i \sin y) + C \\
 &= e^x(x \cos y + i x \sin y + i y \cos y + i^2 y \sin y) + C \\
 &= e^x[(x \cos y - y \sin y) + i(x \sin y + y \cos y)] + C \\
 &= e^x(x \cos y - y \sin y) + i e^x(x \sin y + y \cos y) + C \\
 \therefore u &= e^x(x \cos y - y \sin y) \text{ is the harmonic conjugate}
 \end{aligned}$$

47. Find real part of an analytic function whose imaginary part is

$$v = x^2 + \frac{x}{x^2+y^2} - y^2$$

[M14/CompIT/6M]

Solution:

We have,

$$\begin{aligned}
 v &= x^2 + \frac{x}{x^2+y^2} - y^2 \\
 v_x &= 2x + \frac{(x^2+y^2)[1] - x[2x]}{(x^2+y^2)^2} \\
 v_y &= \frac{(x^2+y^2)[0] - x[2y]}{(x^2+y^2)^2} - 2y
 \end{aligned}$$

Now,

$$f'(z) = u_x + i v_x$$

But $u_x = v_y$

$$f'(z) = v_y + i v_x$$

By Milne Thomson method,

$$\begin{aligned}
 f'(z) &= v_{y(z,0)} + i v_{x(z,0)} \\
 f'(z) &= [0] + i \left[2z + \frac{z^2 - 2z^2}{z^4} \right] \\
 f'(z) &= i \left[2z - \frac{1}{z^2} \right]
 \end{aligned}$$

Integrating, we get

$$\begin{aligned}
 f(z) &= i \left[2 \frac{z^2}{2} + \frac{1}{z} \right] + C \\
 f(z) &= i \left[z^2 + \frac{1}{z} \right] + C
 \end{aligned}$$

Now,

$$\begin{aligned}
 f(z) &= i \left[(x + iy)^2 + \frac{1}{x+iy} \right] + C \\
 &= i \left[x^2 + 2ixy + i^2 y^2 + \frac{1}{x+iy} \times \frac{x-iy}{x-iy} \right] + C \\
 &= i \left[x^2 + 2ixy - y^2 + \frac{x-iy}{x^2-y^2} \right] + C \\
 &= i \left[x^2 + 2ixy - y^2 + \frac{x}{x^2+y^2} - \frac{iy}{x^2+y^2} \right] + C \\
 &= ix^2 + i^2 2xy - iy^2 + \frac{ix}{x^2+y^2} - \frac{i^2 y}{x^2+y^2} + C
 \end{aligned}$$



$$= \left[-2xy + \frac{y}{x^2+y^2} \right] + i \left[x^2 - y^2 + \frac{x}{x^2+y^2} \right] + C$$

$$\therefore u = -2xy + \frac{y}{x^2+y^2}$$

48. If $f(z) = u + iv$ is analytic and $u + v = e^x(\cos y + \sin y)$, find $f(z)$

[M15/AutoMechCivil/6M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } U_x = V_y$$

$$\therefore (1 + i)f'(z) = V_y + iV_x$$

Now,

$$u + v = V = e^x \cos y + e^x \sin y$$

$$V_x = \cos y [e^x] + \sin y [e^x]$$

$$V_y = e^x [-\sin y] + e^x [\cos y]$$

And,

$$(1 + i)f'(z) = V_y + iV_x$$

By Milne – Thomson method,

$$(1 + i)f'(z) = V_{y(z,0)} + iV_{x(z,0)}$$

$$(1 + i)f'(z) = [e^z \cos 0] + i[\cos 0 e^z]$$

$$(1 + i)f'(z) = e^z + ie^z$$

$$(1 + i)f'(z) = (1 + i)e^z$$

$$f'(z) = e^z$$

Integrating w.r.t z , we get,

$$f(z) = e^z + c$$

49. Find the analytic function $f(z) = u + iv$ such that

$$u - v = \frac{\cos x + \sin x - e^{-y}}{2\cos x - e^y - e^{-y}} \quad \text{when } f\left(\frac{\pi}{2}\right) = 0$$

[N13/AutoMechCivil/8M][N17/Biot/6M][N18/MTRX/8M]

[M19/Chem/8M]

Solution:

We have,

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$



Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } V_x = -U_y$$

$$\therefore (1 + i)f'(z) = U_x - iU_y$$

Now,

$$u - v = U = \frac{\cos x + \sin x - e^{-y}}{2\cos x - e^y - e^{-y}} = \frac{\cos x + \sin x - e^{-y}}{2\cos x - 2\cosh y}$$

$$U_x = \frac{(2\cos x - 2\cosh y)[- \sin x + \cos x] - (\cos x + \sin x - e^{-y})[-2\sin x]}{(2\cos x - 2\cosh y)^2}$$

$$U_y = \frac{(2\cos x - 2\cosh y)[e^{-y}] - (\cos x + \sin x - e^{-y})[2\sinh y]}{(2\cos x - 2\cosh y)^2}$$

And,

$$(1 + i)f'(z) = U_x - iU_y$$

By Milne – Thomson method,

$$(1 + i)f'(z) = U_{x(z,0)} - iU_{y(z,0)}$$

$$(1 + i)f'(z) = \left[\frac{(2\cos z - 2)[- \sin z + \cos z] - (\cos z + \sin z - 1)[-2\sin z]}{(2\cos z - 2)^2} \right] - i \left[\frac{(2\cos z - 2)}{(2\cos z - 2)^2} \right]$$

$$(1 + i)f'(z) = \left[\frac{-2\sin z \cos z + 2\cos^2 z + 2\sin z - 2\cos z + 2\sin z \cos z + 2\sin^2 z - 2\sin z}{(2\cos z - 2)^2} \right] - \frac{i}{2\cos z - 2}$$

$$(1 + i)f'(z) = \left[\frac{2 - 2\cos z}{(2\cos z - 2)^2} \right] - \frac{i}{2\cos z - 2}$$

$$(1 + i)f'(z) = \left[-\frac{2\cos z - 2}{(2\cos z - 2)^2} \right] - \frac{i}{2\cos z - 2}$$

$$(1 + i)f'(z) = -\frac{1}{2\cos z - 2} - \frac{i}{2\cos z - 2}$$

$$(1 + i)f'(z) = \frac{-1 - i}{2\cos z - 2}$$

$$(1 + i)f'(z) = \frac{-(1 + i)}{-2(1 - \cos z)}$$

$$f'(z) = \frac{1}{2} \cdot \frac{1}{1 - \cos z}$$

$$f'(z) = \frac{1}{2} \cdot \frac{1}{2 \sin^2 \frac{z}{2}}$$

$$f'(z) = \frac{1}{4} \cdot \operatorname{cosec}^2 \frac{z}{2}$$

Integrating w.r.t z , we get

$$f(z) = \frac{1}{4} \cdot -\frac{\cot \frac{z}{2}}{\frac{1}{2}} + C$$

$$f(z) = -\frac{1}{2} \cdot \cot \frac{z}{2} + C$$

Now, put $z = \frac{\pi}{2}$

$$f\left(\frac{\pi}{2}\right) = -\frac{1}{2} \cdot \cot \frac{\pi}{4} + C$$



$$0 = -\frac{1}{2} \cdot (1) + C$$

$$\therefore C = \frac{1}{2}$$

Thus,

$$f(z) = -\frac{1}{2} \cdot \cot \frac{z}{2} + \frac{1}{2}$$

$$f(z) = \frac{1}{2} \left[1 - \cot \frac{z}{2} \right]$$

50. If $f(z) = u + iv$ is analytic and $u - v = e^x(\cos y - \sin y)$, find $f(z)$
[N14/AutoMechCivil/6M][N16/ElexExtcElectBiomInst/6M]
[N19/Comp/6M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } V_x = -U_y$$

$$\therefore (1 + i)f'(z) = U_x - iU_y$$

Now,

$$u - v = U = e^x \cos y - e^x \sin y$$

$$U_x = \cos y [e^x] - \sin y [e^x]$$

$$U_y = e^x [-\sin y] - e^x [\cos y]$$

And,

$$(1 + i)f'(z) = U_x - iU_y$$

By Milne – Thomson method,

$$(1 + i)f'(z) = U_{x(z,0)} - iU_{y(z,0)}$$

$$(1 + i)f'(z) = [\cos 0 (e^z)] - i[-e^z (\cos 0)]$$

$$(1 + i)f'(z) = e^z + ie^z$$

$$(1 + i)f'(z) = (1 + i)e^z$$

$$f'(z) = e^z$$

Integrating w.r.t z , we get, $f(z) = e^z + c$

51. Find the analytic function $f(z) = u + iv$ in terms of z if

$$u + v = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

[N16/ChemBiot/6M]

Solution:



$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } U_x = V_y$$

$$\therefore (1 + i)f'(z) = V_y + iV_x$$

Now,

$$u + v = V = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$V_x = \frac{(\cosh 2y - \cos 2x)[2\cos 2x] - \sin 2x[2\sin 2x]}{(\cosh 2y - \cos 2x)^2}$$

$$V_y = \frac{(\cosh 2y - \cos 2x)[0] - \sin 2x[2\sinh 2y]}{(\cosh 2y - \cos 2x)^2}$$

And,

$$(1 + i)f'(z) = V_y + iV_x$$

By Milne – Thomson method,

$$(1 + i)f'(z) = V_{y(z,0)} + iV_{x(z,0)}$$

$$(1 + i)f'(z) = [0] + i \left[\frac{(1 - \cos 2z)[2\cos 2z] - \sin 2z[2\sin 2z]}{(1 - \cos 2z)^2} \right]$$

$$(1 + i)f'(z) = i \left[\frac{2\cos 2z - 2\cos^2 2z - 2\sin^2 2z}{(1 - \cos 2z)^2} \right]$$

$$(1 + i)f'(z) = 2i \left[\frac{\cos 2z - 1}{(1 - \cos 2z)^2} \right]$$

$$f'(z) = -\frac{2i}{1+i} \left[\frac{1 - \cos 2z}{(1 - \cos 2z)^2} \right]$$

$$f'(z) = -\frac{2i}{1+i} \left[\frac{1}{1 - \cos 2z} \right]$$

$$f'(z) = -\frac{2i}{1+i} \left[\frac{1}{2\sin^2 z} \right] = -\frac{i}{1+i} \operatorname{cosec}^2 z$$

Integrating w.r.t z , we get,

$$f(z) = -\frac{i}{1+i} \cdot -\cot z + c$$

$$f(z) = \frac{1+i}{2} \cot z + c$$

52. If $u + v = e^{2x}(x\cos 2y - y\sin 2y)$ then find analytic function $f(z)$ by Milne Thomson Method

[M18/Extc/6M]

Solution:

We have,

$$f(z) = u + iv$$



$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } U_x = V_y$$

$$\therefore (1 + i)f'(z) = V_y + iV_x$$

Now,

$$u + v = V = e^{2x}(x\cos 2y - y\sin 2y)$$

$$V_x = \cos 2y[e^{2x}(1) + x(2e^{2x})] - y\sin 2y[2e^{2x}]$$

$$V_y = e^{2x}x[-2\sin 2y] - e^{2x}[y(2\cos 2y) + \sin 2y(1)]$$

$$(1 + i)f'(z) = V_y + iV_x$$

By Milne Thomson method,

$$(1 + i)f'(z) = V_{y(z,0)} + iV_{x(z,0)}$$

$$(1 + i)f'(z) = [e^{2z} + 2ze^{2z}] - i[0]$$

$$f'(z) = \frac{1}{1+i}[e^{2z} + 2ze^{2z}]$$

Integrating, we get

$$f(z) = \frac{1}{1+i}\left[\frac{e^{2z}}{2} + 2z\left(\frac{e^{2z}}{2}\right) - 2\left(\frac{e^{2z}}{4}\right)\right] + C$$

$$f(z) = \frac{1-i}{2}[ze^{2z}] + C$$

53. Find the analytic function $f(z) = u + iv$ in terms of z if

$$u + v = \frac{2\sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x}$$

[M15/ChemBiot/8M][N15/CompIT/6M][N18/Elect/8M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } U_x = V_y$$

$$\therefore (1 + i)f'(z) = V_y + iV_x$$

Now,



$$u + v = V = \frac{2\sin 2x}{e^{2y} + e^{-2y} - 2\cos 2x} = \frac{2\sin 2x}{2\cosh 2y - 2\cos 2x} = \frac{\sin 2x}{\cosh 2y - \cos 2x}$$

$$V_x = \frac{(\cosh 2y - \cos 2x)[2\cos 2x] - \sin 2x[2\sin 2x]}{(\cosh 2y - \cos 2x)^2}$$

$$V_y = \frac{(\cosh 2y - \cos 2x)[0] - \sin 2x[2\sinh 2y]}{(\cosh 2y - \cos 2x)^2}$$

And,

$$(1 + i)f'(z) = V_y + iV_x$$

By Milne – Thomson method,

$$(1 + i)f'(z) = V_{y(z,0)} + iV_{x(z,0)}$$

$$(1 + i)f'(z) = [0] + i \left[\frac{(1 - \cos 2z)[2\cos 2z] - \sin 2z[2\sin 2z]}{(1 - \cos 2z)^2} \right]$$

$$(1 + i)f'(z) = i \left[\frac{2\cos 2z - 2\cos^2 2z - 2\sin^2 2z}{(1 - \cos 2z)^2} \right]$$

$$(1 + i)f'(z) = 2i \left[\frac{\cos 2z - 1}{(1 - \cos 2z)^2} \right]$$

$$f'(z) = -\frac{2i}{1+i} \left[\frac{1 - \cos 2z}{(1 - \cos 2z)^2} \right]$$

$$f'(z) = -\frac{2i}{1+i} \left[\frac{1}{1 - \cos 2z} \right]$$

$$f'(z) = -\frac{2i}{1+i} \left[\frac{1}{2\sin^2 z} \right] = -\frac{i}{1+i} \operatorname{cosec}^2 z$$

Integrating w.r.t z, we get,

$$f(z) = -\frac{i}{1+i} \cdot -\cot z + c$$

$$f(z) = \frac{1+i}{2} \cot z + c$$

54. Find the analytic function $f(z) = u + iv$ in terms of z if

$$u - v = (x - y)(x^2 + 4xy + y^2)$$

[M19/IT/6M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x, we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } V_x = -U_y$$

$$\therefore (1 + i)f'(z) = U_x - iU_y$$

Now,

$$u - v = U = (x - y)(x^2 + 4xy + y^2)$$

$$U_x = (x - y)[2x + 4y] + (x^2 + 4xy + y^2)[1]$$



$$U_y = (x - y)[4x + 2y] + (x^2 + 4xy + y^2)[-1]$$

And,

$$(1 + i)f'(z) = U_x - iU_y$$

By Milne – Thomson method,

$$(1 + i)f'(z) = U_{x(z,0)} - iU_{y(z,0)}$$

$$(1 + i)f'(z) = [(z)(2z) + (z^2)] - i[(z)(4z) + (z^2)(-1)]$$

$$(1 + i)f'(z) = 3z^2 - i(3z^2)$$

$$(1 + i)f'(z) = (1 - i)(3z^2)$$

$$f'(z) = \frac{1-i}{1+i} \cdot 3z^2 = -i \cdot 3z^2$$

Integrating w.r.t z , we get,

$$f(z) = -iz^3 + C$$

55. If $f(z) = u + iv$ is an analytic function of $z = x + iy$ and $u + v = \cos x \cosh y - \sin x \sinh y$, find $f(z)$ in terms of z [N18/Elex/8M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } U_x = V_y$$

$$\therefore (1 + i)f'(z) = V_y + iV_x$$

Now,

$$u + v = V = \cos x \cosh y - \sin x \sinh y$$

$$V_x = \cosh y [-\sin x] - \sinh y [\cos x]$$

$$V_y = \cos x [\sinh y] - \sin x [\cosh y]$$

And,

$$(1 + i)f'(z) = V_y + iV_x$$

By Milne – Thomson method,

$$(1 + i)f'(z) = V_{y(z,0)} + iV_{x(z,0)}$$

$$(1 + i)f'(z) = [0 - \sin z] + i[-\sin z - 0]$$

$$(1 + i)f'(z) = -\sin z - i\sin z$$

$$f'(z) = -\frac{(1+i)}{1+i} \cdot \sin z$$

Integrating w.r.t z , we get,

$$f(z) = -(-\cos z) + c = \cos z + c$$



56. Find the analytic function $f(z) = u + iv$ in terms of z if

$$u + v = \frac{x}{x^2 + y^2}$$

[M15/CompIT/6M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$

Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } U_x = V_y$$

$$\therefore (1 + i)f'(z) = V_y + iV_x$$

Now,

$$u + v = V = \frac{x}{x^2 + y^2}$$

$$V_x = \frac{(x^2 + y^2)[1] - x[2x]}{(x^2 + y^2)^2}$$

$$V_y = \frac{(x^2 + y^2)[0] - x[2y]}{(x^2 + y^2)^2}$$

And,

$$(1 + i)f'(z) = V_y + iV_x$$

By Milne – Thomson method,

$$(1 + i)f'(z) = V_{y(z,0)} + iV_{x(z,0)}$$

$$(1 + i)f'(z) = [0] + i \left[\frac{z^2 - 2z^2}{z^4} \right]$$

$$(1 + i)f'(z) = -\frac{i}{z^2}$$

$$f'(z) = -\frac{i}{1+i} \cdot \frac{1}{z^2}$$

Integrating w.r.t z , we get,

$$f(z) = -\frac{i}{1+i} \cdot -\frac{1}{z} + c$$

$$f(z) = \frac{1}{1+i} \left(\frac{i}{z} \right) + c$$

57. Find the analytic function $f(z) = u + iv$ in terms of z if

$$u - v = \frac{(x-y)}{(x^2 + 4xy + y^2)}$$

[N15/ElexExtcElectBiomInst/6M]

Solution:

$$f(z) = u + iv$$

$$if(z) = iu + i^2v$$



Adding, we get

$$f(z) + if(z) = u + iv + iu + i^2v$$

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$(1 + i)f(z) = U + iV$$

Diff w.r.t x , we get

$$(1 + i)f'(z) = U_x + iV_x$$

$$\text{But } V_x = -U_y$$

$$\therefore (1 + i)f'(z) = U_x - iU_y$$

Now,

$$u - v = U = \frac{(x-y)}{(x^2+4xy+y^2)}$$

$$U_x = \frac{(x^2+4xy+y^2)[1] - (x-y)[2x+4y]}{(x^2+4xy+y^2)^2}$$

$$U_y = \frac{(x^2+4xy+y^2)[-1] - (x-y)[4x+2y]}{(x^2+4xy+y^2)^2}$$

And,

$$(1 + i)f'(z) = U_x - iU_y$$

By Milne – Thomson method,

$$(1 + i)f'(z) = U_{x(z,0)} - iU_{y(z,0)}$$

$$(1 + i)f'(z) = \left[\frac{z^2 - 2z^2}{(z^2)^2} \right] - i \left[\frac{-z^2 - 4z^2}{(z^2)^2} \right]$$

$$(1 + i)f'(z) = -\frac{1}{z^2} + \frac{5i}{z^2}$$

$$(1 + i)f'(z) = \frac{-1+5i}{z^2}$$

$$f'(z) = \frac{-1+5i}{1+i} \cdot \frac{1}{z^2}$$

Integrating w.r.t z , we get,

$$f(z) = \frac{-1+5i}{1+i} \cdot \frac{1}{z} + C$$

$$f(z) = -\frac{(2+3i)}{z} + C$$

58. Determine the analytic function $f(z)$ given $3u + 2v = y^2 - x^2 + 16xy$
[N13/CompIT/6M][M19/Elect/8M]

Solution:

We have,

$$3u + 2v = y^2 - x^2 + 16xy \dots\dots\dots(1)$$

Diff partially eqn (1) w.r.t. x , we get

$$3u_x + 2v_x = -2x + 16y \dots\dots\dots(2)$$

Diff partially eqn (1) w.r.t. y , we get

$$3u_y + 2v_y = 2y + 16x$$

$$\text{But } u_y = -v_x \text{ \& } v_y = u_x$$

$$-3v_x + 2u_x = 2y + 16x$$



$$2u_x - 3v_x = 16x + 2y \dots\dots\dots(3)$$

Multiplying eqn (2) by 2 and eqn (3) by 3 and subtracting, we get

$$4v_x + 9v_x = -4x + 32y - 48x - 6y$$

$$13v_x = -52x + 26y$$

$$\therefore v_x = -4x + 2y$$

Multiplying eqn (2) by 3 and eqn (3) by 2 and adding, we get

$$9u_x + 4u_x = -6x + 48y + 32x + 4y$$

$$13u_x = 26x + 52y$$

$$\therefore u_x = 2x + 4y$$

Now,

$$f'(z) = u_x + iv_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + iv_{x(z,0)}$$

$$f'(z) = [2z] + i[-4z]$$

$$f'(z) = 2z(1 - 2i)$$

Integrating, we get

$$f(z) = 2 \left[\frac{z^2}{2} \right] (1 - 2i) + C$$

$$f(z) = (1 - 2i)z^2 + C$$

59. Find the analytic function $f(z) = u + iv$ in terms of z if

$$u - 3v = x^2 - y^2 - 5x + y + 2$$

[N19/Elect/6M]

Solution:

We have,

$$u - 3v = x^2 - y^2 - 5x + y + 2 \dots\dots\dots(1)$$

Diff partially eqn (1) w.r.t. x , we get

$$u_x - 3v_x = 2x - 5 \dots\dots\dots(2)$$

Diff partially eqn (1) w.r.t. y , we get

$$u_y - 3v_y = -2y + 1$$

But $u_y = -v_x$ & $v_y = u_x$

$$-v_x - 3u_x = -2y + 1$$

$$3u_x + v_x = 2y - 1 \dots\dots\dots(3)$$

Multiplying eqn (2) by 3 and subtracting with eqn (3), we get

$$-10v_x = 6x - 14 - 2y$$

$$v_x = -\frac{3}{5}x + \frac{y}{5} + \frac{7}{5}$$

Multiplying eqn (3) by 3 and adding with eqn (2), we get

$$10u_x = 2x + 6y - 8$$

$$u_x = \frac{x}{5} + \frac{3y}{5} - \frac{4}{5}$$

Now,



$$f'(z) = u_x + iv_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + iv_{x(z,0)}$$

$$f'(z) = \left[\frac{z}{5} - \frac{4}{5} \right] + i \left[-\frac{3z}{5} + \frac{7}{5} \right]$$

Integrating, we get

$$f(z) = \frac{1}{5} \left[\frac{z^2}{2} - 4z \right] - \frac{i}{5} \left[\frac{3z^2}{2} - 7z \right] + C$$

60. Find the analytic function $f(z) = u + iv$ in terms of z if

$$3u - 7v = x^3 + x^2 - 3xy^2 - y^2 - 3yx^2 + y^3 - 2xy$$

[N17/Elect/6M][M19/Inst/6M]

Solution:

We have,

$$3u - 7v = x^3 + x^2 - 3xy^2 - y^2 - 3yx^2 + y^3 - 2xy \dots\dots\dots(1)$$

Diff partially eqn (1) w.r.t. x , we get

$$3u_x - 7v_x = 3x^2 + 2x - 3y^2 - 6xy - 2y \dots\dots\dots(2)$$

Diff partially eqn (1) w.r.t. y , we get

$$3u_y - 7v_y = -6xy - 2y - 3x^2 + 3y^2 - 2x$$

But $u_y = -v_x$ & $v_y = u_x$

$$-3v_x - 7u_x = -3x^2 - 2x + 3y^2 - 6xy - 2y$$

$$7u_x + 3v_x = 3x^2 + 2x - 3y^2 + 6xy + 2y \dots\dots\dots(3)$$

Multiplying eqn (2) by 7 and eqn (3) by 3 and subtracting, we get

$$-49v_x - 9v_x = 12x^2 + 8x - 12y^2 - 60xy - 20y$$

$$-58v_x = 12x^2 + 8x - 12y^2 - 60xy - 20y$$

$$\therefore v_x = \frac{12x^2 + 8x - 12y^2 - 60xy - 20y}{-58}$$

Multiplying eqn (2) by 3 and eqn (3) by 7 and adding, we get

$$9u_x + 49u_x = 30x^2 + 20x - 30y^2 + 24xy + 8y$$

$$58u_x = 30x^2 + 20x - 30y^2 + 24xy + 8y$$

$$\therefore u_x = \frac{30x^2 + 20x - 30y^2 + 24xy + 8y}{58}$$

Now,

$$f'(z) = u_x + iv_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + iv_{x(z,0)}$$

$$f'(z) = \left[\frac{30z^2 + 20z}{58} \right] + i \left[\frac{12z^2 + 8z}{-58} \right]$$

$$f'(z) = \frac{1}{58} [(30z^2 + 20z) - i(12z^2 + 8z)]$$

Integrating, we get

$$f(z) = \frac{1}{58} \left[\left(30 \frac{z^3}{3} + 20 \frac{z^2}{2} \right) - i \left(12 \frac{z^3}{3} + 8 \frac{z^2}{2} \right) \right] + C$$



$$f(z) = \frac{1}{58} [(10z^3 + 10z^2 - 4iz^3 - 4iz^2)] + C$$

$$f(z) = \frac{10-4i}{58} (z^3 + z^2) + C$$

61. Find the analytic function $f(z) = u + iv$ in terms of z if

$$7u - 4v = x^3 + x^2 - 3xy^2 - y^2 - 3yx^2 + y^3 - 2xy$$

[N18/Inst/6M]

Solution:

We have,

$$7u - 4v = x^3 + x^2 - 3xy^2 - y^2 - 3yx^2 + y^3 - 2xy \dots\dots\dots(1)$$

Diff partially eqn (1) w.r.t. x , we get

$$7u_x - 4v_x = 3x^2 + 2x - 3y^2 - 6xy - 2y \dots\dots\dots(2)$$

Diff partially eqn (1) w.r.t. y , we get

$$7u_y - 4v_y = -6xy - 2y - 3x^2 + 3y^2 - 2x$$

But $u_y = -v_x$ & $v_y = u_x$

$$-7v_x - 4u_x = -3x^2 - 2x + 3y^2 - 6xy - 2y$$

$$4u_x + 7v_x = 3x^2 + 2x - 3y^2 + 6xy + 2y \dots\dots\dots(3)$$

Multiplying eqn (2) by 4 and eqn (3) by 7 and subtracting, we get

$$-16v_x - 49v_x = -9x^2 - 6x + 9y^2 - 66xy - 22y$$

$$-65v_x = -9x^2 - 6x + 9y^2 - 66xy - 22y$$

$$\therefore v_x = \frac{-9x^2 - 6x + 9y^2 - 66xy - 22y}{-65}$$

Multiplying eqn (2) by 7 and eqn (3) by 4 and adding, we get

$$49u_x + 16u_x = 33x^2 + 22x - 33y^2 - 18xy - 6y$$

$$65u_x = 33x^2 + 22x - 33y^2 - 18xy - 6y$$

$$\therefore u_x = \frac{33x^2 + 22x - 33y^2 - 18xy - 6y}{65}$$

Now,

$$f'(z) = u_x + iv_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + iv_{x(z,0)}$$

$$f'(z) = \left[\frac{33z^2 + 22z}{65} \right] + i \left[\frac{-9z^2 - 6z}{-65} \right]$$

$$f'(z) = \frac{1}{65} [(33z^2 + 22z) + i(9z^2 + 6z)]$$

Integrating, we get

$$f(z) = \frac{1}{65} \left[\left(33 \frac{z^3}{3} + 22 \frac{z^2}{2} \right) + i \left(9 \frac{z^3}{3} + 6 \frac{z^2}{2} \right) \right] + C$$

$$f(z) = \frac{1}{65} [(11z^3 + 11z^2 + 3iz^3 + 3iz^2)] + C$$

$$f(z) = \frac{11+3i}{65} (z^3 + z^2) + C$$

62. Find the analytic function $f(z) = u + iv$ in terms of z if

$$4u - 5v = x^3 + x^2 - 3xy^2 - y^2 - 3yx^2 + y^3 - 2xy$$

[N17/M18/Biom/6M]

Solution:

We have,

$$4u - 5v = x^3 + x^2 - 3xy^2 - y^2 - 3yx^2 + y^3 - 2xy \dots\dots\dots(1)$$

Diff partially eqn (1) w.r.t. x , we get

$$4u_x - 5v_x = 3x^2 + 2x - 3y^2 - 6xy - 2y \dots\dots\dots(2)$$

Diff partially eqn (1) w.r.t. y , we get

$$4u_y - 5v_y = -6xy - 2y - 3x^2 + 3y^2 - 2x$$

But $u_y = -v_x$ & $v_y = u_x$

$$-4v_x - 5u_x = -3x^2 - 2x + 3y^2 - 6xy - 2y$$

$$5u_x + 4v_x = 3x^2 + 2x - 3y^2 + 6xy + 2y \dots\dots\dots(3)$$

Multiplying eqn (2) by 5 and eqn (3) by 4 and subtracting, we get

$$-25v_x - 16v_x = 3x^2 + 2x - 3y^2 - 54xy - 18y$$

$$-41v_x = 3x^2 + 2x - 3y^2 - 54xy - 18y$$

$$\therefore v_x = \frac{3x^2 + 2x - 3y^2 - 54xy - 18y}{-41}$$

Multiplying eqn (2) by 4 and eqn (3) by 5 and adding, we get

$$16u_x + 25u_x = 27x^2 + 18x - 27y^2 + 6xy + 2y$$

$$41u_x = 27x^2 + 18x - 27y^2 + 6xy + 2y$$

$$\therefore u_x = \frac{27x^2 + 18x - 27y^2 + 6xy + 2y}{41}$$

Now,

$$f'(z) = u_x + iv_x$$

By Milne Thomson method,

$$f'(z) = u_{x(z,0)} + iv_{x(z,0)}$$

$$f'(z) = \left[\frac{27z^2 + 18z}{41} \right] + i \left[\frac{3z^2 + 2z}{-41} \right]$$

$$f'(z) = \frac{1}{41} [(27z^2 + 18z) - i(3z^2 + 2z)]$$

Integrating, we get

$$f(z) = \frac{1}{41} \left[\left(27 \frac{z^3}{3} + 18 \frac{z^2}{2} \right) - i \left(3 \frac{z^3}{3} + 2 \frac{z^2}{2} \right) \right] + C$$

$$f(z) = \frac{1}{41} [9z^3 + 9z^2 - iz^3 - iz^2] + C$$

$$f(z) = \frac{9-i}{41} (z^3 + z^2) + C$$

Type IV: Orthogonal Curves

Find the orthogonal trajectory of the family of curves

1. $x^3y - xy^3 = c$

[M14/AutoMechCivil/6M][M15/ChemBiot/6M]

[M17/AutoMechCivil/6M][N17/Biot/6M][M19/MTRX/6M]

[N19/Chem/6M]

Solution:

Let $u = x^3y - xy^3$

$$u_x = y[3x^2] - y^3[1]$$

$$u_y = x^3[1] - x[3y^2]$$

Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = [0 - 0] - i[z^3 - 0]$$

$$f'(z) = -i z^3$$

Integrating, we get

$$f(z) = -i \frac{z^4}{4} + C$$

$$f(z) = -\frac{i(x+iy)^4}{4} + C$$

$$f(z) = -\frac{i(x^4 + 4x^3iy + 6x^2i^2y^2 + 4xi^3y^3 + i^4y^4)}{4} + C$$

$$f(z) = -\frac{ix^4}{4} - i^2x^3y - \frac{i^36x^2y^2}{4} - i^4xy^3 - \frac{i^5y^4}{4} + C$$

$$f(z) = (x^3y - xy^3) + i\left(-\frac{x^4}{4} + \frac{6x^2y^2}{4} - \frac{y^4}{4}\right) + C$$

$$\therefore v = -\frac{x^4}{4} + \frac{6x^2y^2}{4} - \frac{y^4}{4}$$

Let $\frac{3x^2y^2}{2} - \frac{y^4}{4} - \frac{x^4}{4} = c$

$$\therefore x^4 + y^4 - 6x^2y^2 = -4c$$

$\therefore x^4 - 6x^2y^2 + y^4 = c'$ becomes the orthogonal trajectory of the family of curves.

2. $e^{-x}\cos y + xy = \alpha$

[N13/CompIT/6M][N13/Biot/6M][M16/AutoMechCivil/5M]

[N17/Elex/5M][M18/Biot/6M][N18/Comp/5M][N19/IT/6M]

Solution:

Let $u = e^{-x}\cos y + xy$

$$u_x = \cos y[-e^{-x}] + y[1]$$

$$u_y = e^{-x}[-\sin y] + x[1]$$



Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = [\cos 0(-e^{-z}) + 0] - i[0 + z]$$

$$f'(z) = -e^{-z} - iz$$

Integrating, we get

$$f(z) = -\left[\frac{e^{-z}}{-1}\right] - i\left[\frac{z^2}{2}\right] + C$$

$$f(z) = e^{-z} - \frac{iz^2}{2} + C$$

$$f(z) = e^{-(x+iy)} - \frac{i(x+iy)^2}{2} + C$$

$$f(z) = e^{-x} \cdot e^{-iy} - \frac{i}{2}(x^2 + 2ixy + i^2y^2) + C$$

$$f(z) = e^{-x}(\cos y - i \sin y) - \frac{ix^2}{2} - i^2xy - \frac{i^3y^2}{2} + C$$

$$f(z) = (e^{-x} \cos y + xy) + i\left(-e^{-x} \sin y + \frac{x^2}{2} + \frac{y^2}{2}\right) + C$$

$$\therefore v = -e^{-x} \sin y + \frac{x^2}{2} + \frac{y^2}{2}$$

Let $-e^{-x} \sin y + \frac{x^2}{2} + \frac{y^2}{2} = c$

Thus, $-e^{-x} \sin y + \frac{x^2}{2} + \frac{y^2}{2} = c$ becomes the orthogonal trajectory of the family of curves.

3. $r^2 \cos 2\theta = \alpha$ Ans. $r^2 \sin 2\theta = \beta$

4. $3x^2y - y^3 = \text{constant}$

[M15/CompIT/6M][M17/CompIT/6M][N17/MTRX/6M]

Solution:

Let $u = 3x^2y - y^3$

$$u_x = 3y[2x]$$

$$u_y = 3x^2[1] - [3y^2]$$

Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = [0] - i[3z^2]$$

$$f'(z) = -3iz^2$$

Integrating, we get

$$f(z) = -3i\left[\frac{z^3}{3}\right] + C$$



$$\begin{aligned}
 f(z) &= -iz^3 + C \\
 f(z) &= -i(x + iy)^3 + C \\
 f(z) &= -i(x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + C \\
 f(z) &= -ix^3 - 3i^2x^2y - 3i^3xy^2 - i^4y^3 + C \\
 f(z) &= (3x^2y - y^3) + i(-x^3 + 3xy^2) + C \\
 \therefore v &= -x^3 + 3xy^2
 \end{aligned}$$

Thus, $-x^3 + 3xy^2 = c$ becomes the orthogonal trajectory of the family of curves.

5. $e^x \cos y - xy = c$

[N15/ChemBiot/6M][M18/Extc/5M][N18/Extc/6M][M18/Chem/6M]

Solution:

Let $u = e^x \cos y - xy$

$$u_x = \cos y [e^x] - y[1]$$

$$u_y = e^x [-\sin y] - x[1]$$

Now,

$$f'(z) = u_x + iv_x$$

But $v_x = -u_y$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = [\cos 0(e^z) + 0] - i[0 - z]$$

$$f'(z) = e^z + iz$$

Integrating, we get

$$f(z) = [e^z] + i \left[\frac{z^2}{2} \right] + C$$

$$f(z) = e^z + \frac{iz^2}{2} + C$$

$$f(z) = e^{(x+iy)} + \frac{i(x+iy)^2}{2} + C$$

$$f(z) = e^x \cdot e^{iy} + \frac{i}{2}(x^2 + 2ixy + i^2y^2) + C$$

$$f(z) = e^x(\cos y + isiny) + \frac{ix^2}{2} + i^2xy + \frac{i^3y^2}{2} + C$$

$$f(z) = (e^x \cos y - xy) + i \left(e^x \sin y + \frac{x^2}{2} - \frac{y^2}{2} \right) + C$$

$$\therefore v = e^x \sin y + \frac{x^2}{2} - \frac{y^2}{2}$$

Let $e^x \sin y + \frac{x^2}{2} - \frac{y^2}{2} = c$

Thus, $e^x \sin y + \frac{x^2}{2} - \frac{y^2}{2} = c$ becomes the orthogonal trajectory of the family of curves.

6. $e^{-x}(x \sin y - y \cos y) = c$

Ans. $e^{-x}(x \cos y + y \sin y) = c$



7. $2x - x^3 + 3xy^2 = a$
[M16/ChemBiot/6M][N16/AutoMechCivil/6M][M18/N18/MTRX/6M]

Solution:

$$\begin{aligned}\text{Let } u &= 2x - x^3 + 3xy^2 \\ u_x &= 2[1] - [3x^2] + 3y^2[1] \\ u_y &= 3x[2y]\end{aligned}$$

Now,

$$f'(z) = u_x + iv_x$$

$$\text{But } v_x = -u_y$$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,

$$f'(z) = [2 - 3z^2] - i[0]$$

$$f'(z) = 2 - 3z^2$$

Integrating, we get

$$f(z) = 2z - 3\left[\frac{z^3}{3}\right] + C$$

$$f(z) = 2z - z^3 + C$$

$$f(z) = 2(x + iy) - (x + iy)^3 + C$$

$$f(z) = 2x + i2y - (x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + C$$

$$f(z) = 2x + i2y - x^3 - 3ix^2y + 3xy^2 + iy^3 + C$$

$$f(z) = (2x - x^3 + 3xy^2) + i(2y - 3x^2y + y^3) + C$$

$$\therefore v = 2y - 3x^2y + y^3$$

Let $2y - 3x^2y + y^3 = c$ becomes the orthogonal trajectory of the family of curves.

8. $u = \text{constant}$ where $u = x^2 - y^2 - 2xy + 2x - 3y$
Ans. $2xy + 2y + x^2 - y^2 + 3y = c$
9. $u = \text{constant}$ where $u = x^2 - y^2 + 5x + y - \frac{y}{x^2 + y^2}$
Ans. $2xy + 5y + x + \frac{x}{x^2 + y^2} = c$
10. $3x^2y + 2x^2 - y^3 - 2y^2 = c$
[M14/ElexExtcElectBiomInst/5M]

Solution:

$$\text{Let } u = 3x^2y + 2x^2 - y^3 - 2y^2$$

$$u_x = 3y[2x] + 2[2x]$$

$$u_y = 3x^2[1] - [3y^2] - 2[2y]$$

Now,

$$f'(z) = u_x + iv_x$$

$$\text{But } v_x = -u_y$$

$$f'(z) = u_x - iu_y$$

By Milne Thomson method,



$$f'(z) = [0 + 4z] - i[3z^2]$$

$$f'(z) = 4z - 3iz^2$$

Integrating, we get

$$f(z) = 4 \left[\frac{z^2}{2} \right] - 3i \left[\frac{z^3}{3} \right] + C$$

$$f(z) = 2z^2 - iz^3 + C$$

$$f(z) = 2(x + iy)^2 - i(x + iy)^3 + C$$

$$f(z) = 2(x^2 + 2ixy + i^2y^2) - i(x^3 + 3x^2iy + 3xi^2y^2 + i^3y^3) + C$$

$$f(z) = 2x^2 + 4ixy - 2y^2 - ix^3 - 3i^2x^2y - 3i^3xy^2 - i^4y^3 + C$$

$$f(z) = (2x^2 - 2y^2 + 3x^2y - y^3) + i(4xy - x^3 + 3xy^2) + C$$

$$\therefore v = 4xy - x^3 + 3xy^2$$

Thus, $4xy - x^3 + 3xy^2 = c$ becomes the orthogonal trajectory of the family of curves.